Generating Natural Language Proofs with Verifier-Guided Search

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PRINCETON UNIVERSITY

PRINCETON NLP

Reasoning in Natural Language

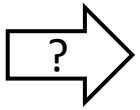
homes are buildings

energy is used for heating buildings

solar is a kind of energy

solar is renewable

Assumptions



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Conclusion

Reasoning in Natural Language

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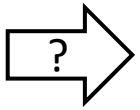

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Conclusion

- Studied extensively in automated theorem proving
- **Remains challenging in natural language**
  - Fuzzy, imprecise, requiring implicit knowledge
  - No well-defined inference rules

Reasoning in Natural Language

[Clark et al. IJCAI 2020]

LLMs for "soft" reasoning over natural language

homes are buildings

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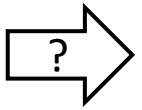

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Task: Proof Generation

[Dalvi et al. EMNLP 2021]

Hypothesis (h):

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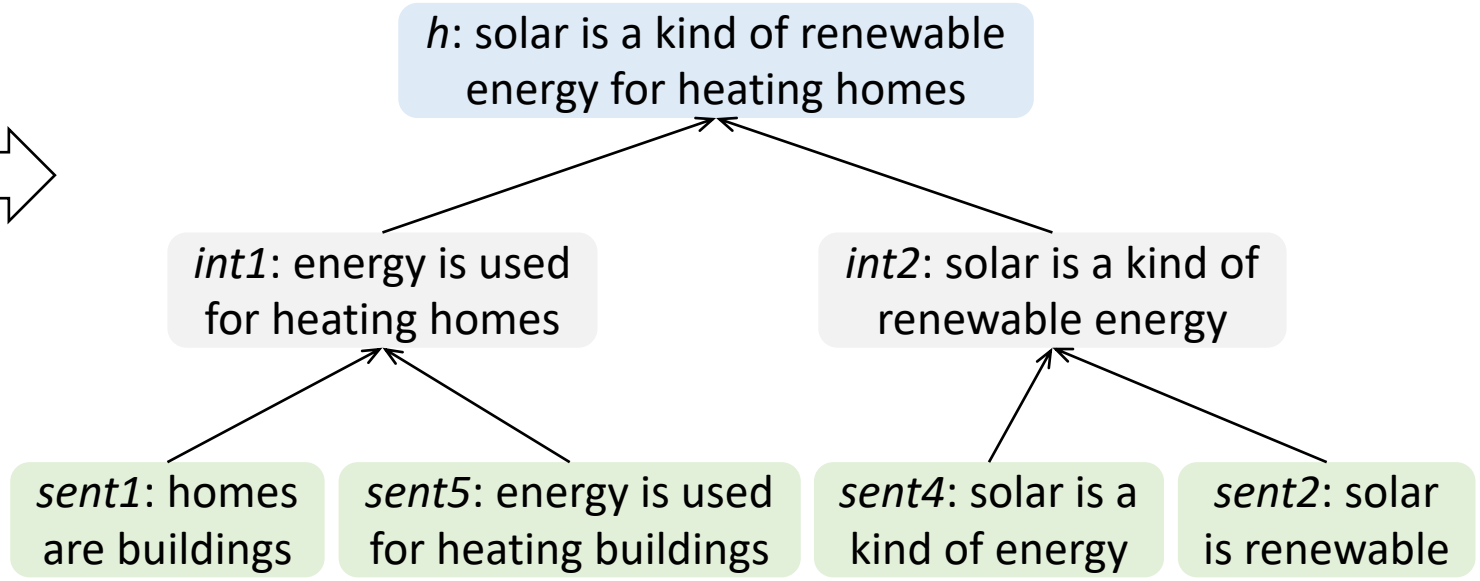
Proof tree (T):



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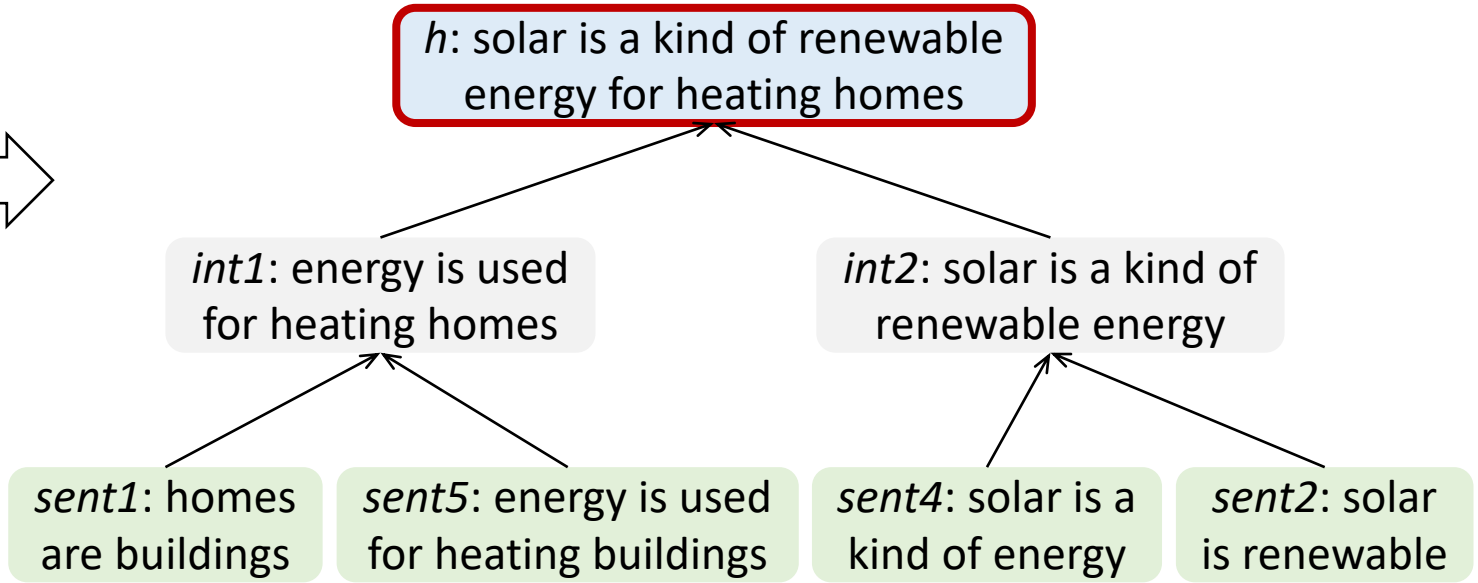

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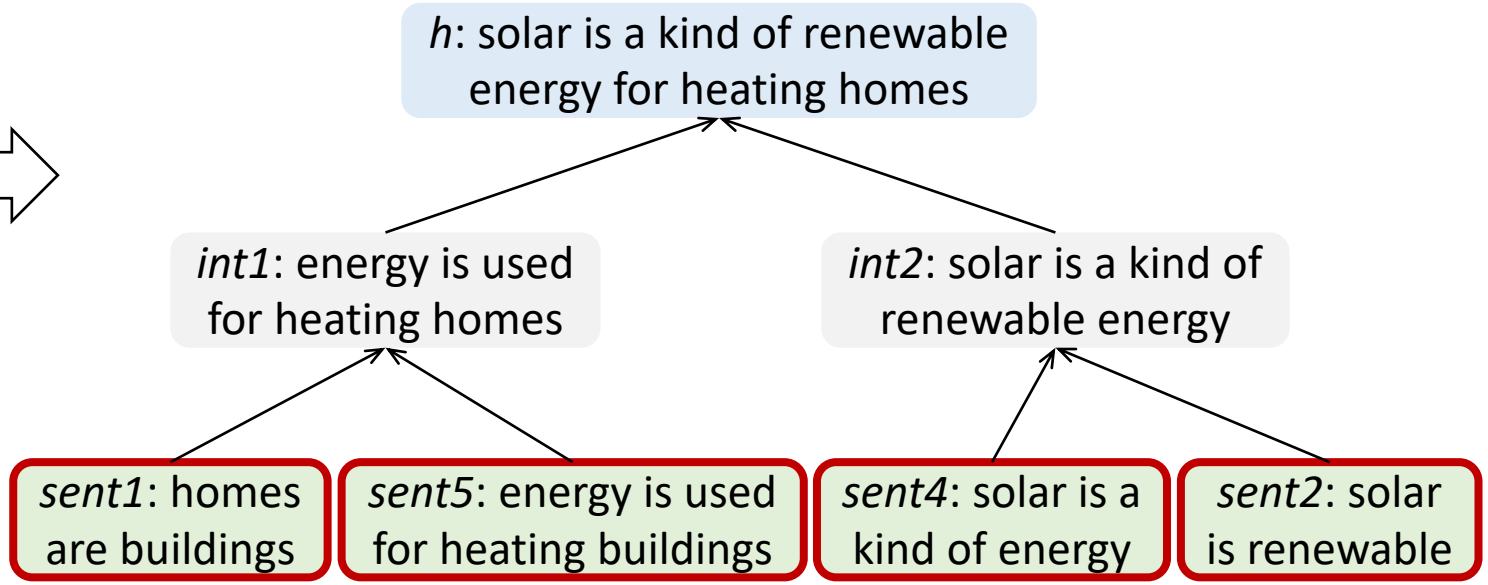

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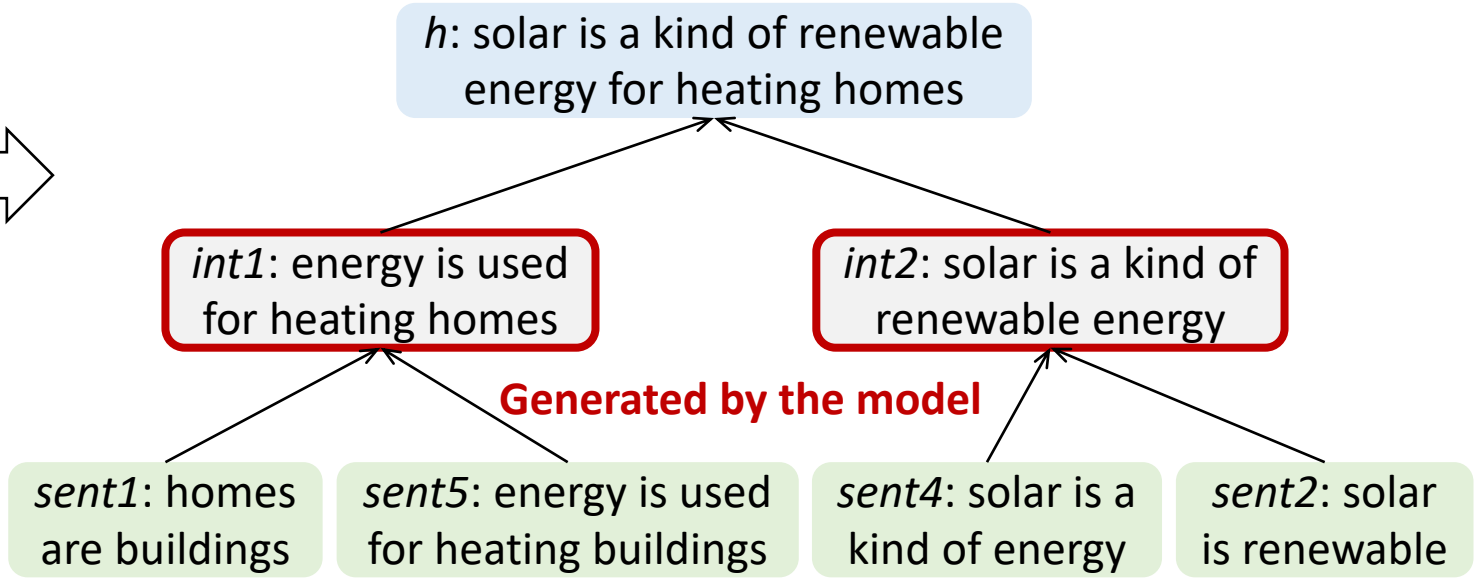

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Single-Shot Methods

Generate the entire proof altogether

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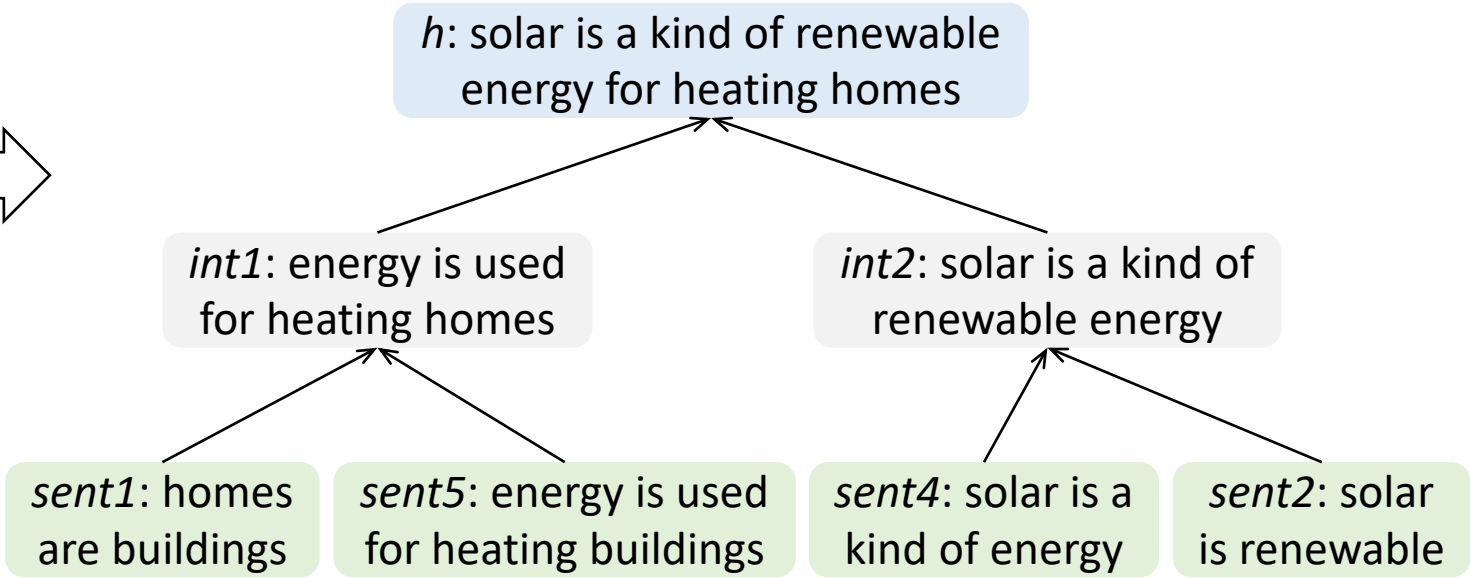

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Stepwise Methods

Generate the proof step by step

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Proof tree (T):

Step 1

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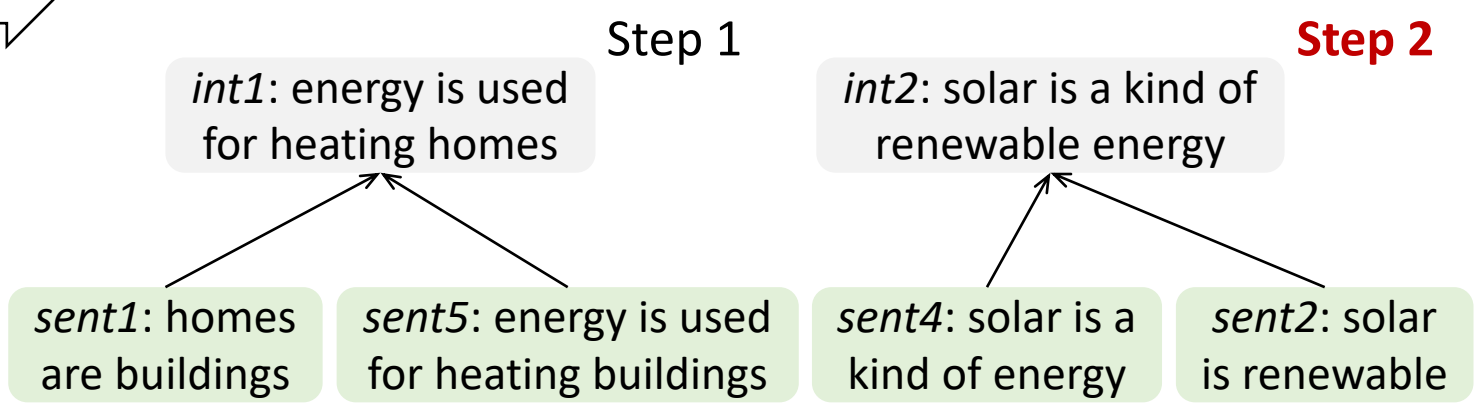

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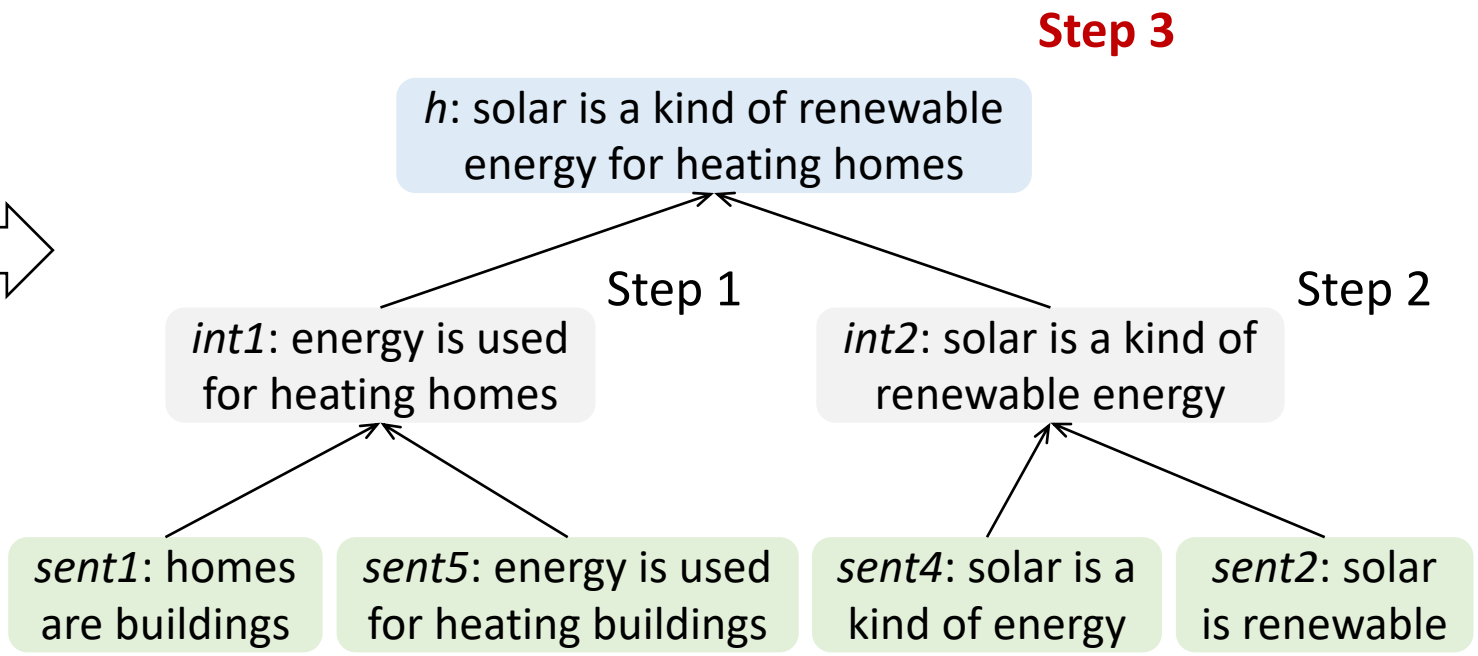

Output

Single-Shot vs. Stepwise Methods

Generate the entire proof altogether

- PRover [Saha et al. EMNLP 2020]
- EntailmentWriter [Dalvi et al. EMNLP 2021]
- PRobr [Sun et al. Findings of ACL 2021]

Generate the proof step by step

- ProofWriter [Tafjord et al. Findings of ACL 2021]
- FaiRR [Sanyal et al. ACL 2022]
- SCSearch [Bostrom et al. Findings of EMNLP 2022]
- MetGen [Hong et al. Findings of NAACL 2022]

✓ Can better leverage compositionality and generalize to longer proofs

✗ Achieved limited success on challenging proofs authored by humans (e.g., EntailmentBank)

Stepwise Methods

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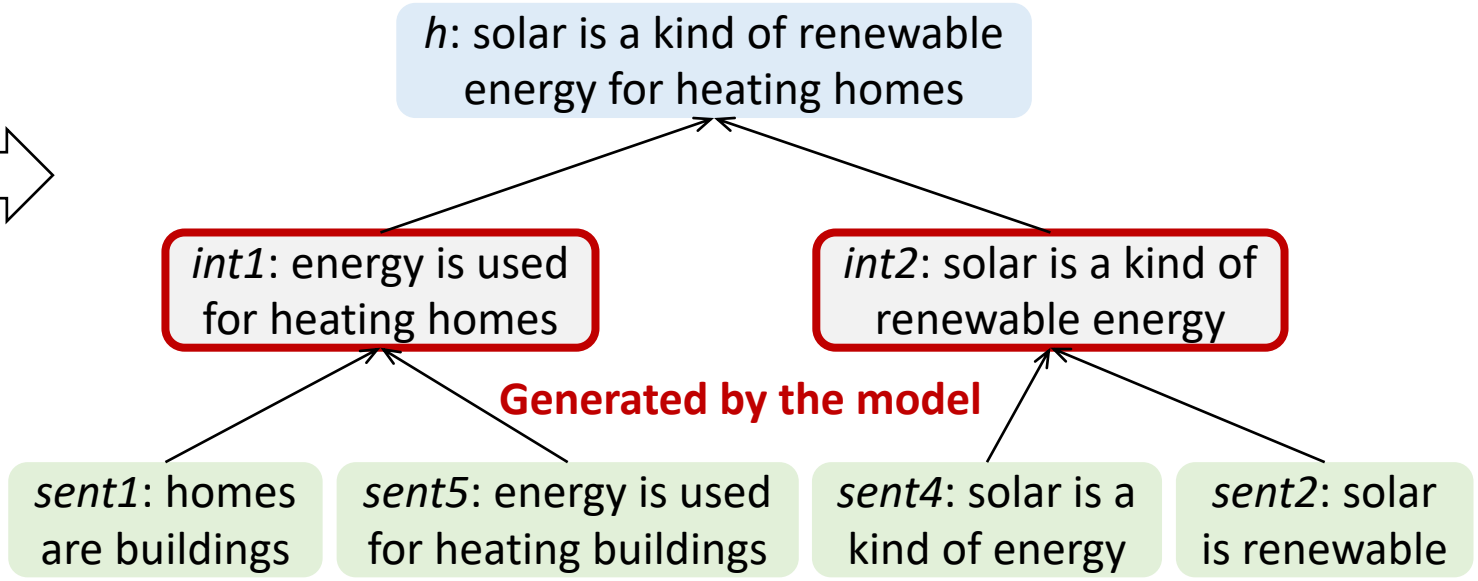

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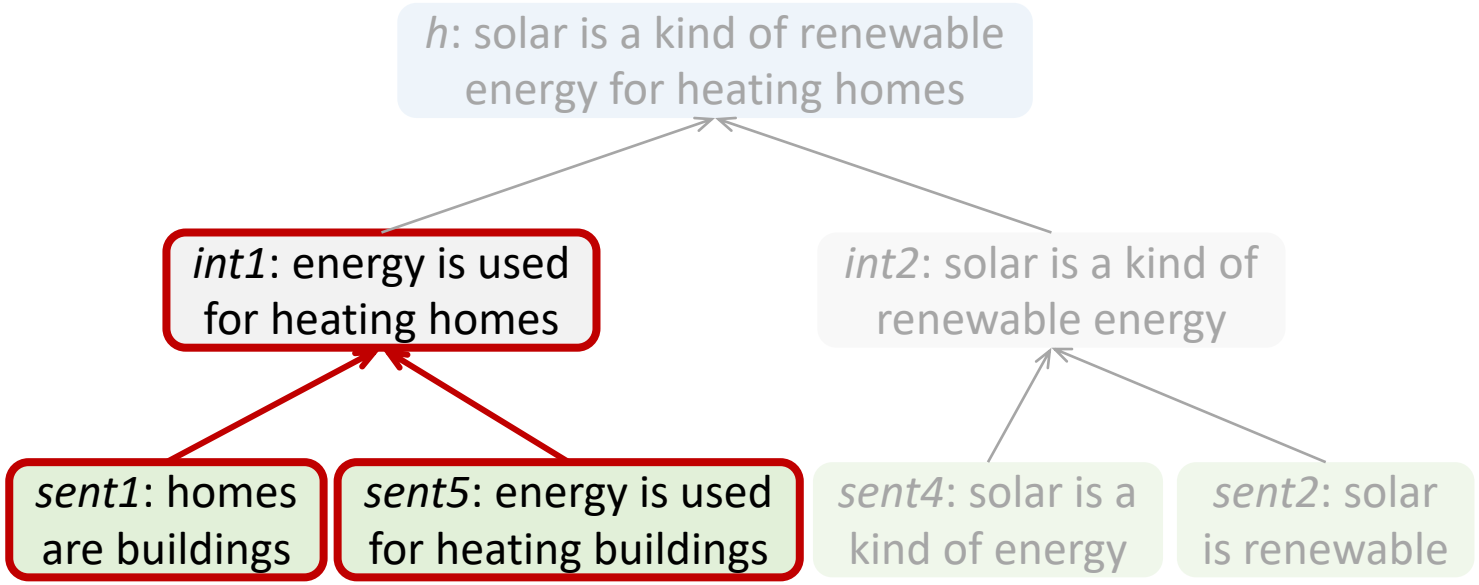
Challenges in Generating Valid and Relevant Steps

- Many valid steps are irrelevant (not useful for proving the hypothesis)



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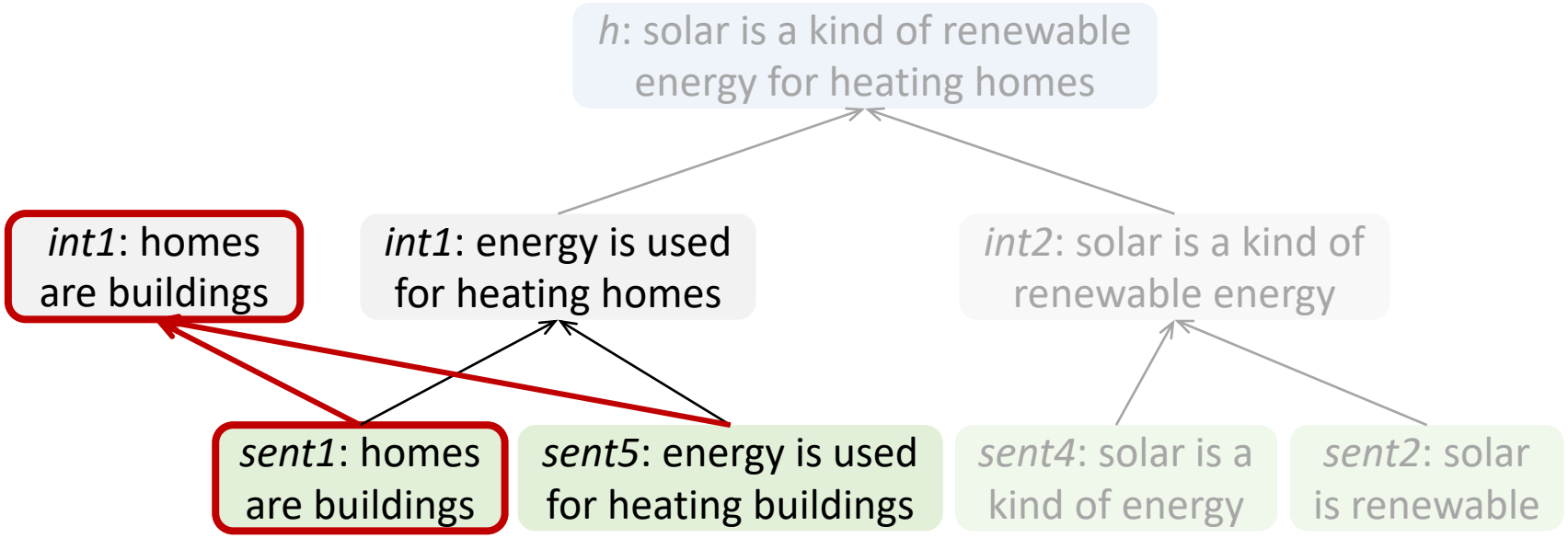

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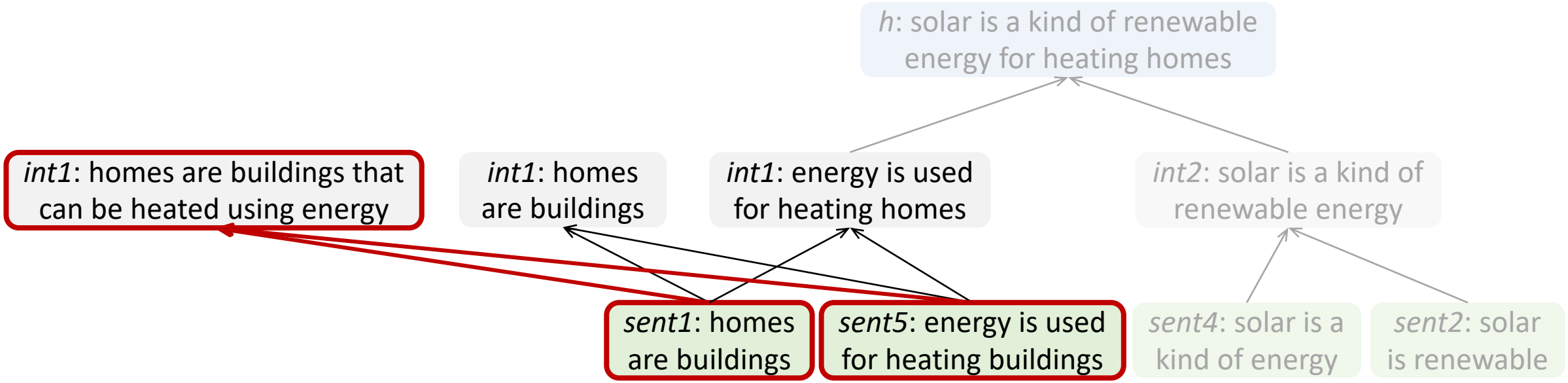

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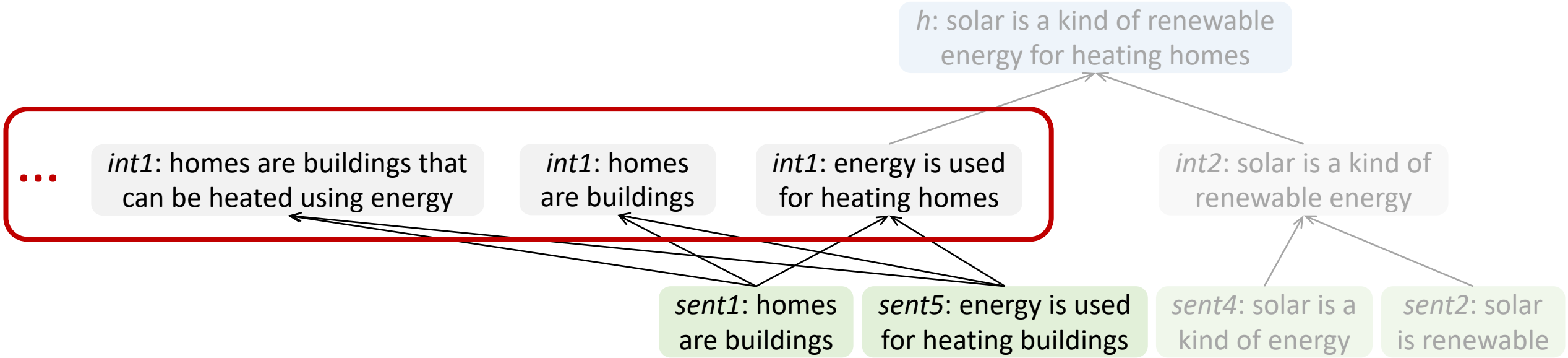

Challenges in Generating Valid and Relevant Steps

- Many valid steps are irrelevant (not useful for proving the hypothesis)
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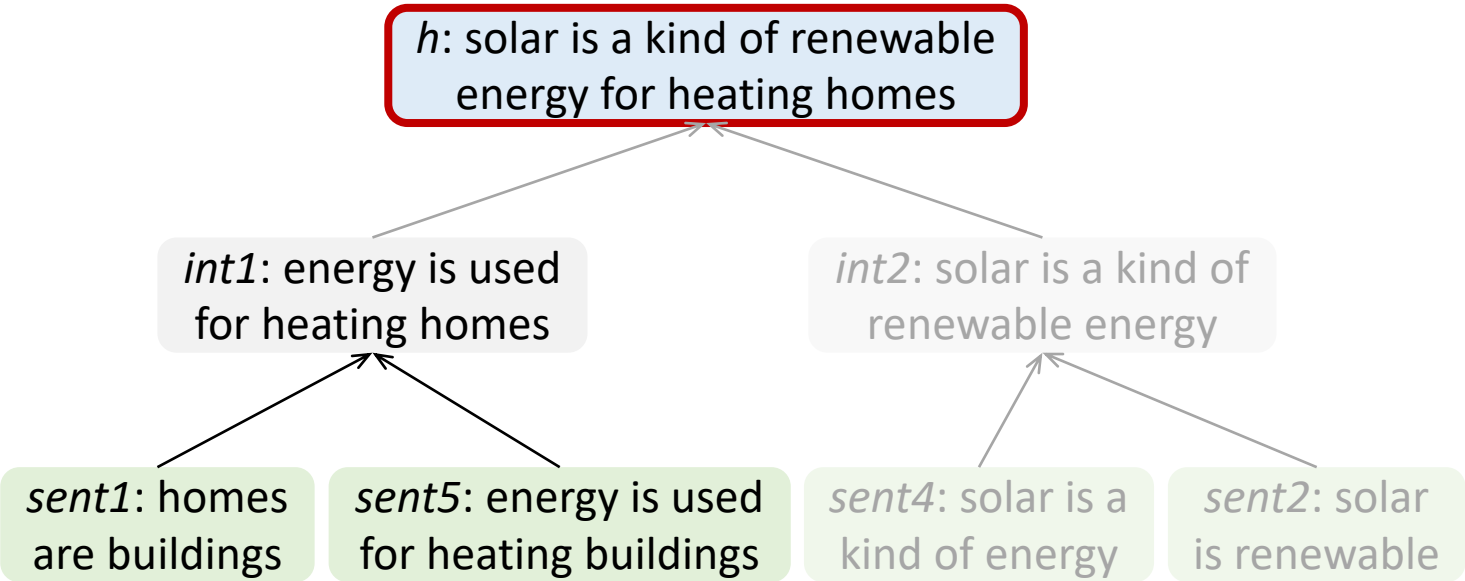
[Sanyal et al. ACL 2022]
[Bostrom et al. arXiv 2022]



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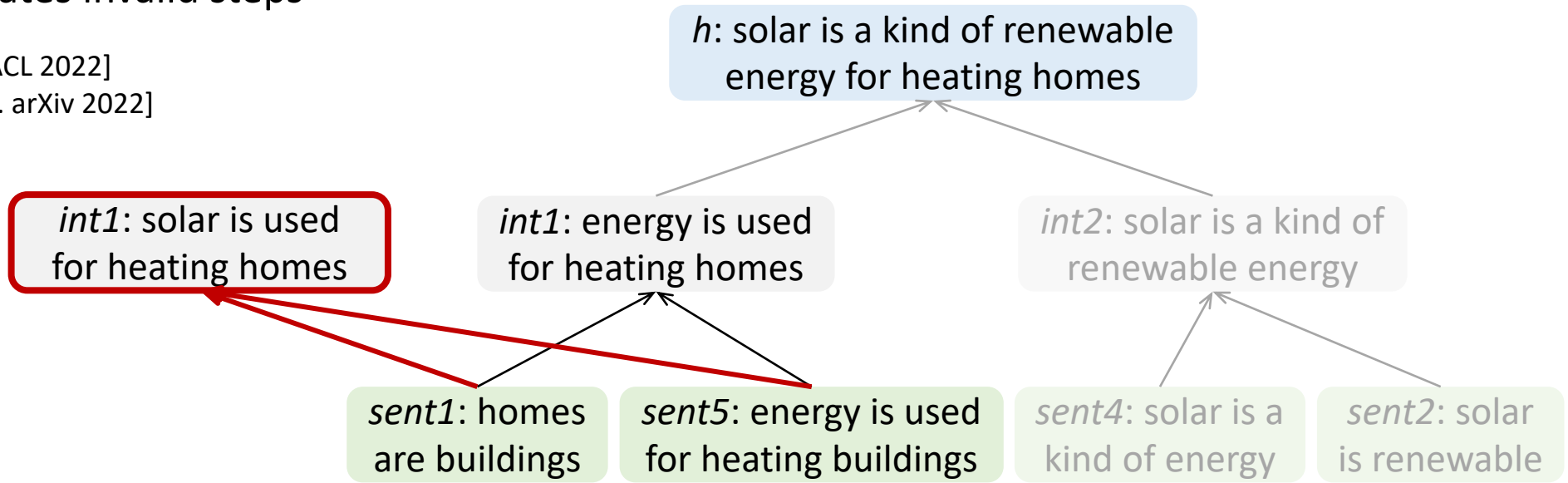

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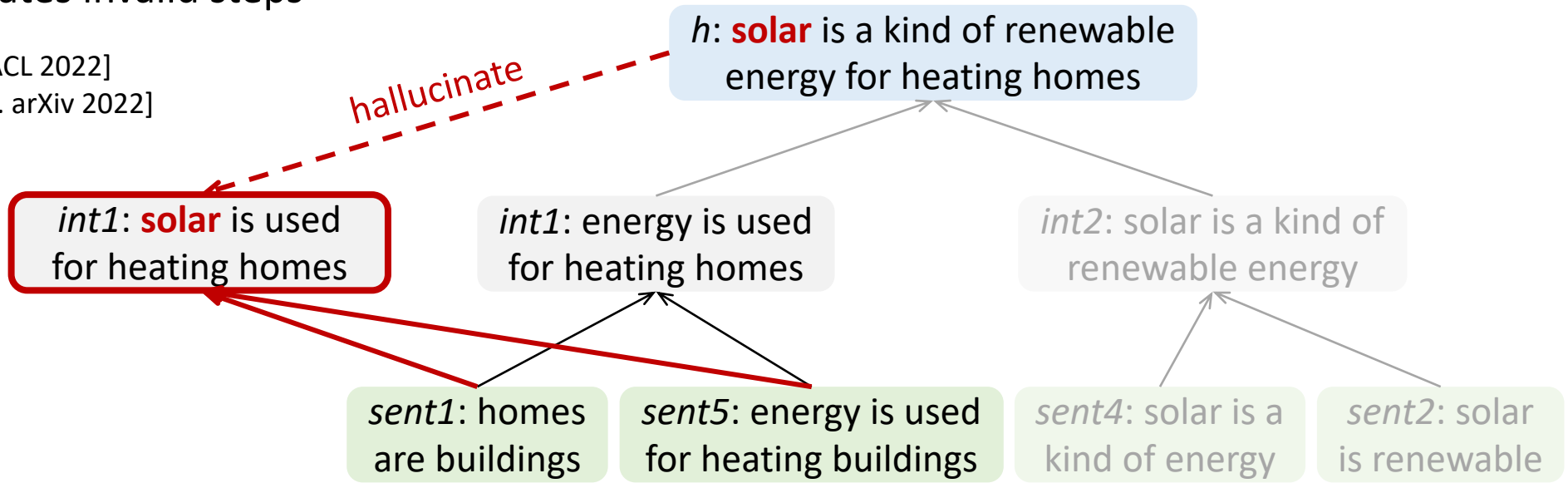

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  - **Struggle to generate valid and relevant steps**
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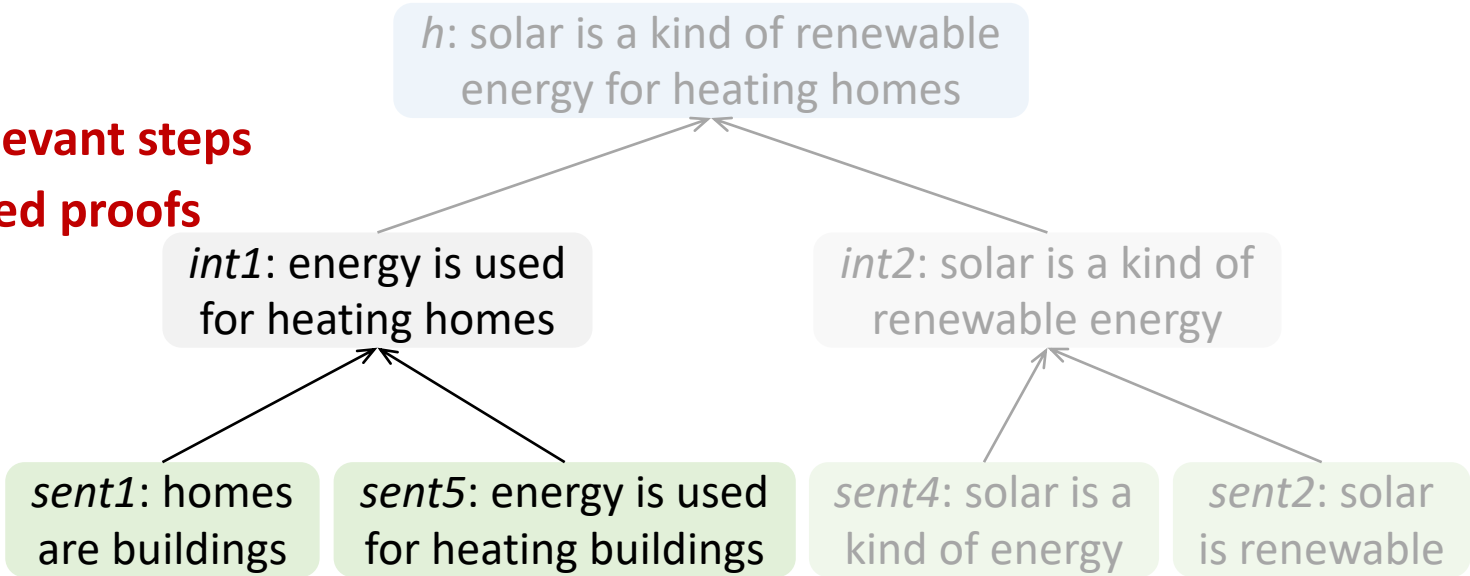

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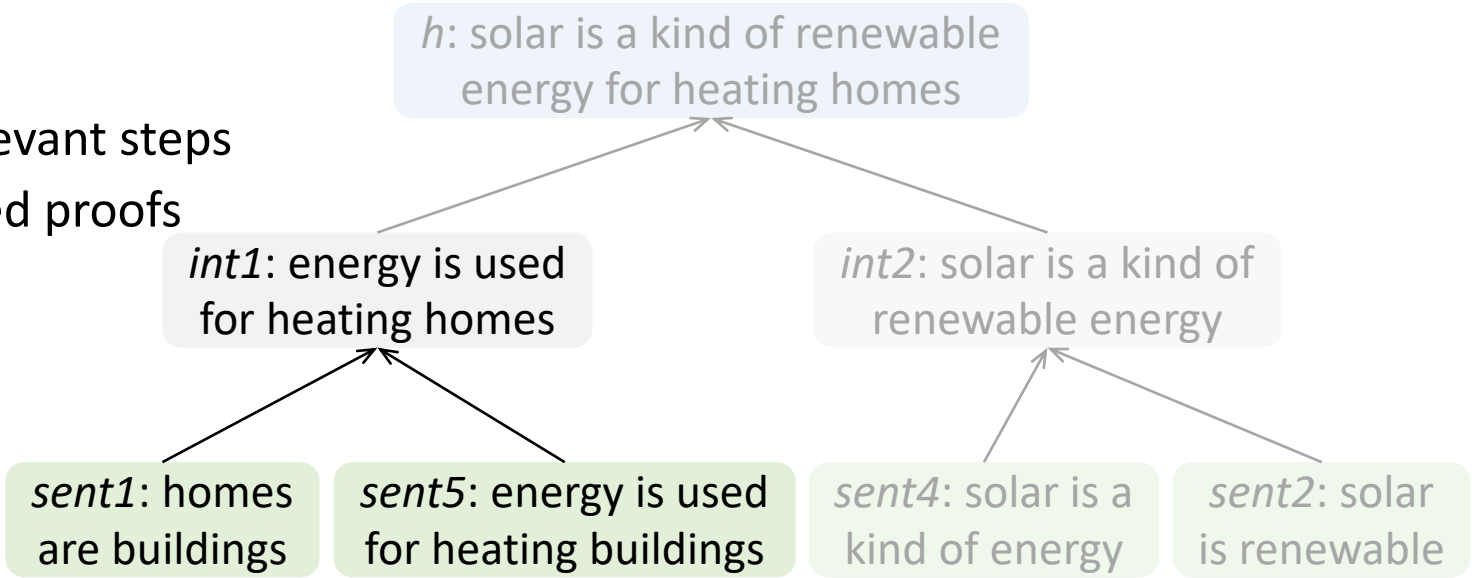
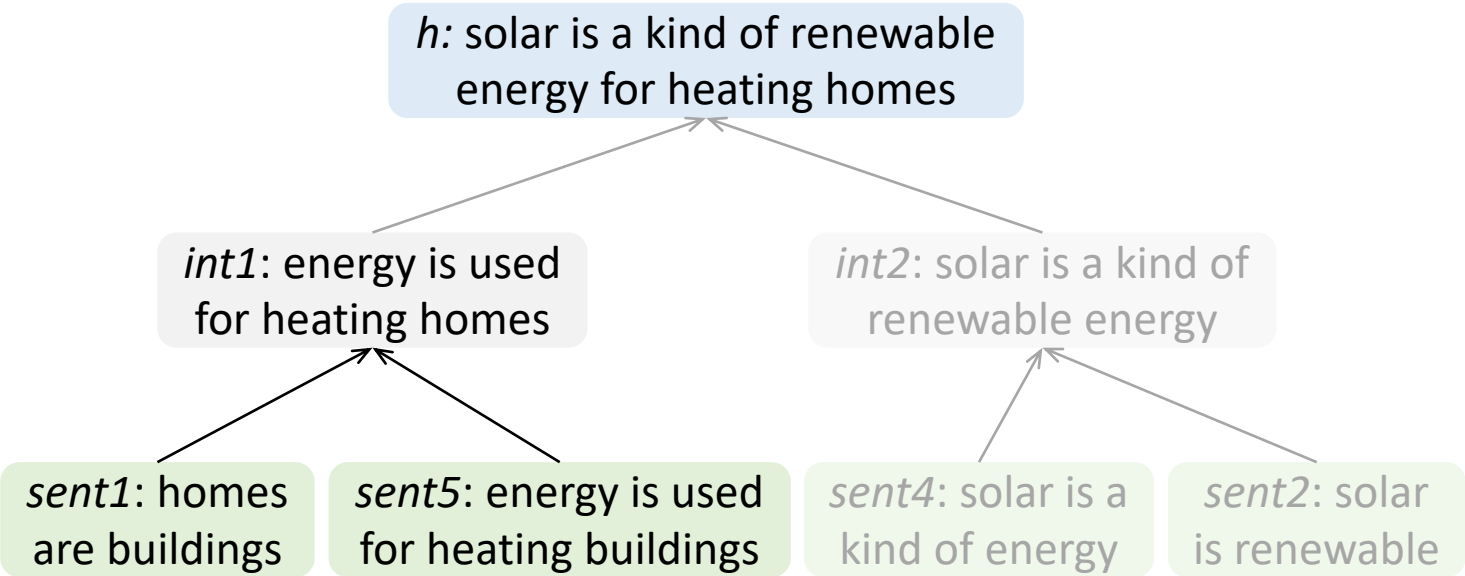


- **Our solution: a new method for stepwise proof generation**

NLProofS: Natural Language Proof Search

- A new method for **stepwise proof generation**



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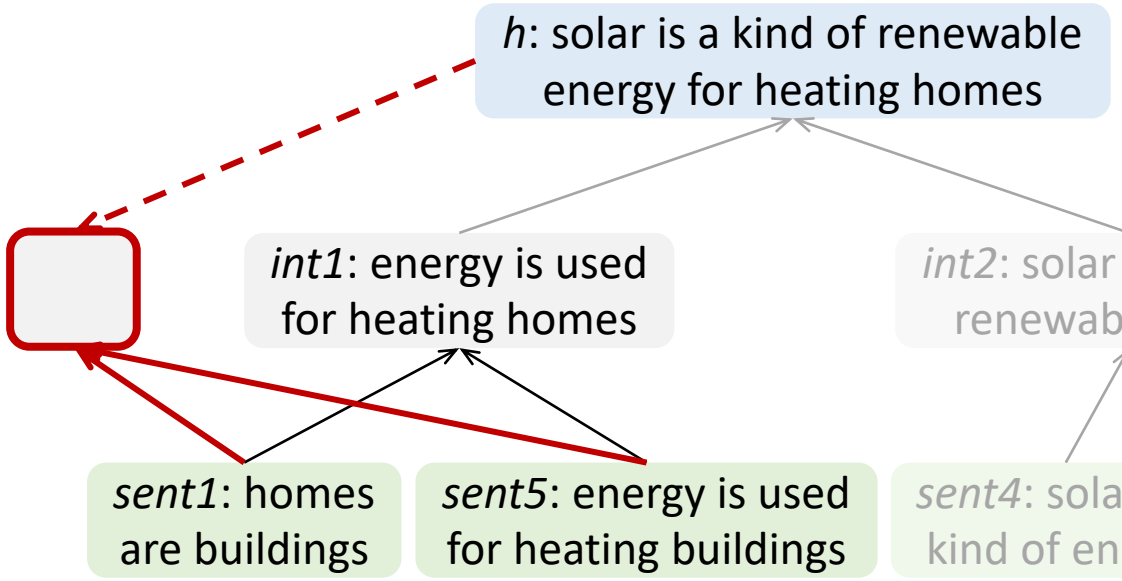
Prover

Generate candidate proof steps



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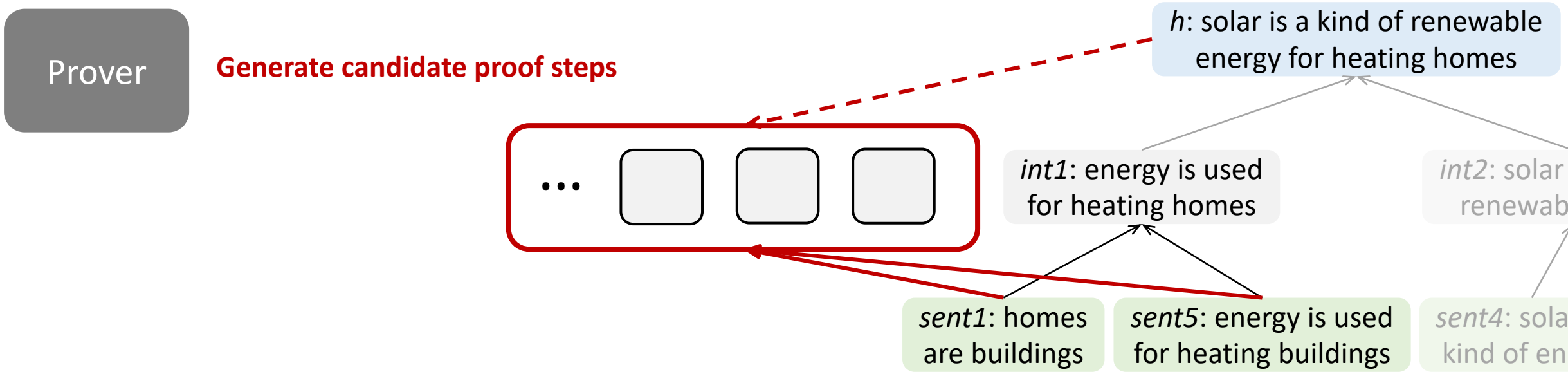

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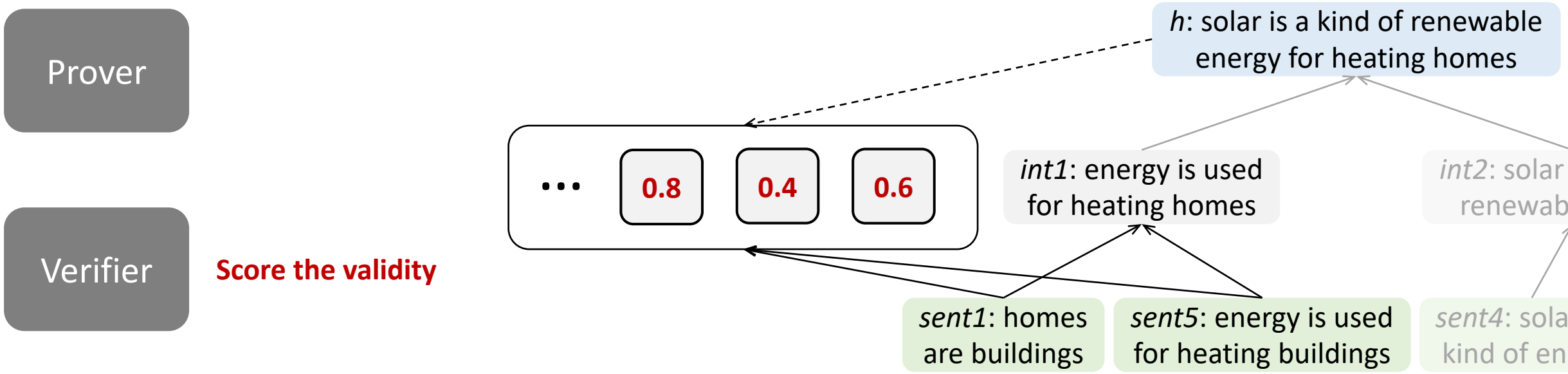

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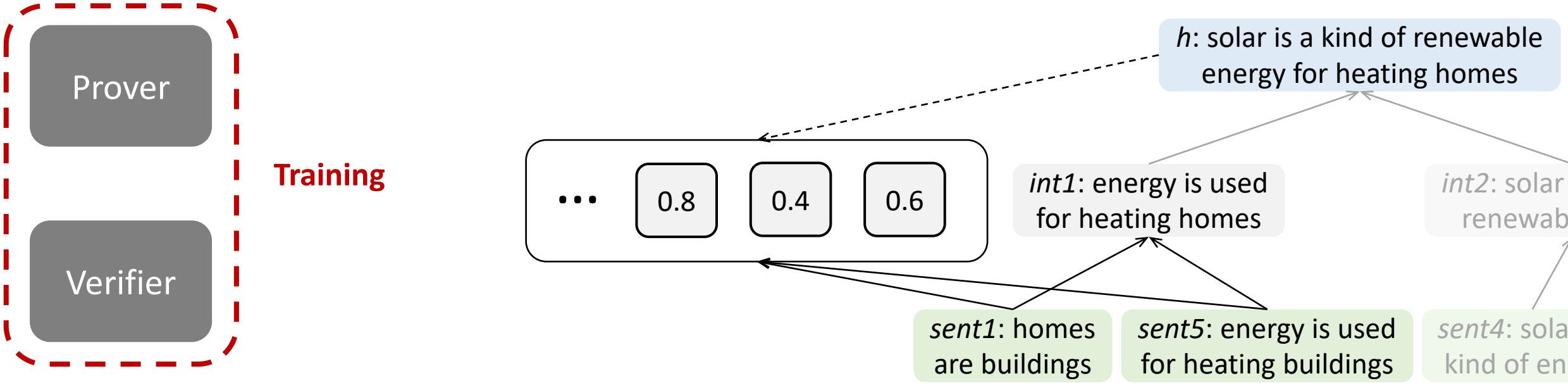

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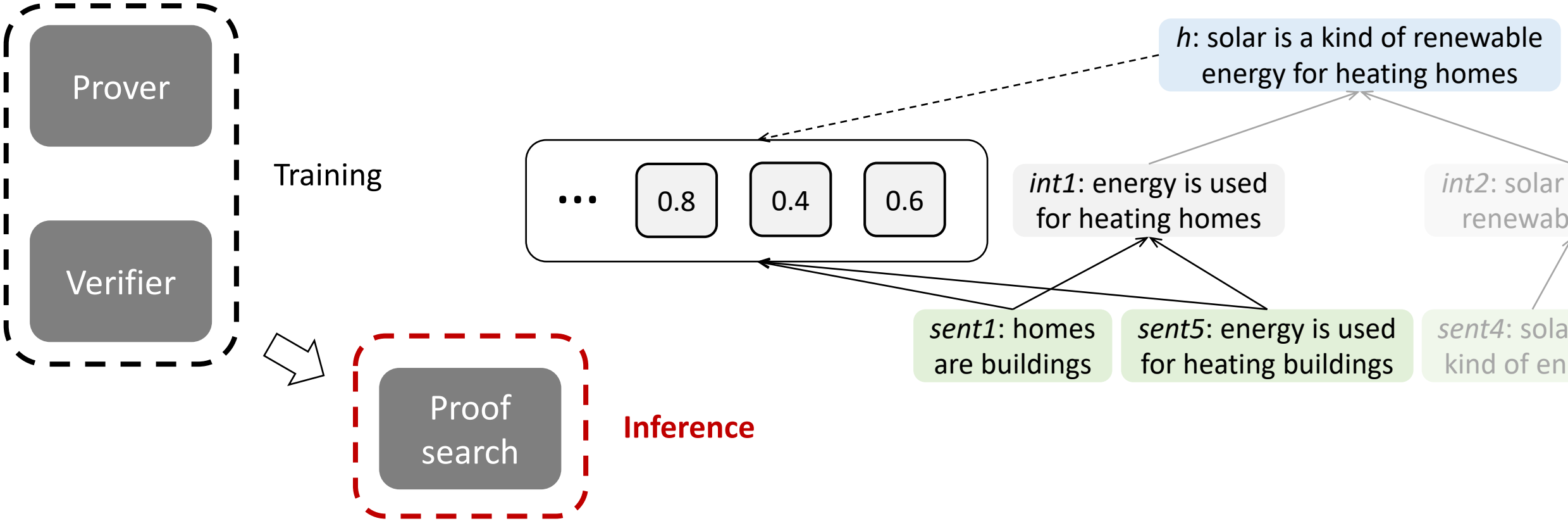

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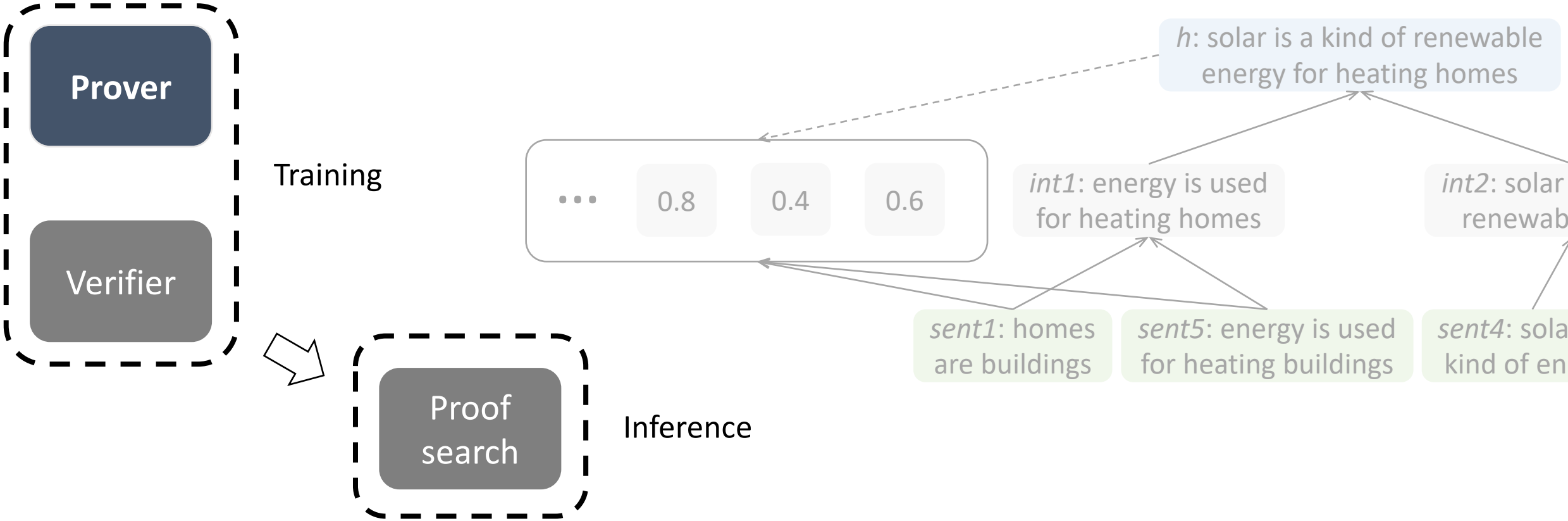

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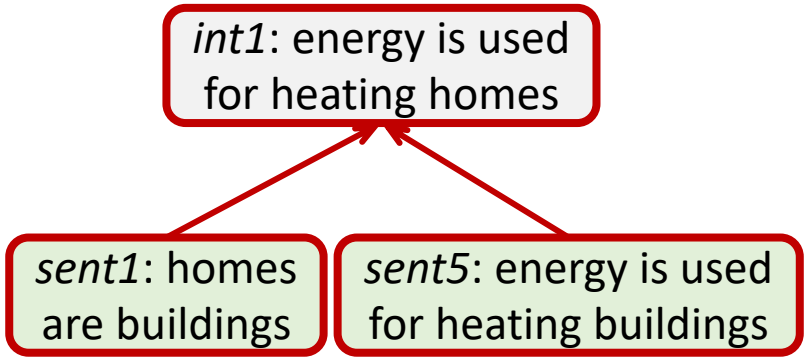
1-step partial proof



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1-step partial proof

int1: energy is used for heating homes

sent1: homes are buildings

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The 2nd step

int2: solar is a kind of renewable energy

sent4: solar is a kind of energy

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Stepwise Prover

[Raffle et al. JMLR 2020] [Tafjord et al. Findings of ACL 2021]

- **Finetune a T5 model to predict the next proof step**

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Encode input/output as text sequences

$hypothesis$ = solar is a kind of renewable energy for heating homes ;
$facts$ = sent1: homes are buildings sent2: solar is renewable … ;
$partial_proof$ = sent1 & sent5 -> int1: energy is used for heating homes;

T5

sent2 & sent4 -> int2: solar is a kind of renewable energy;

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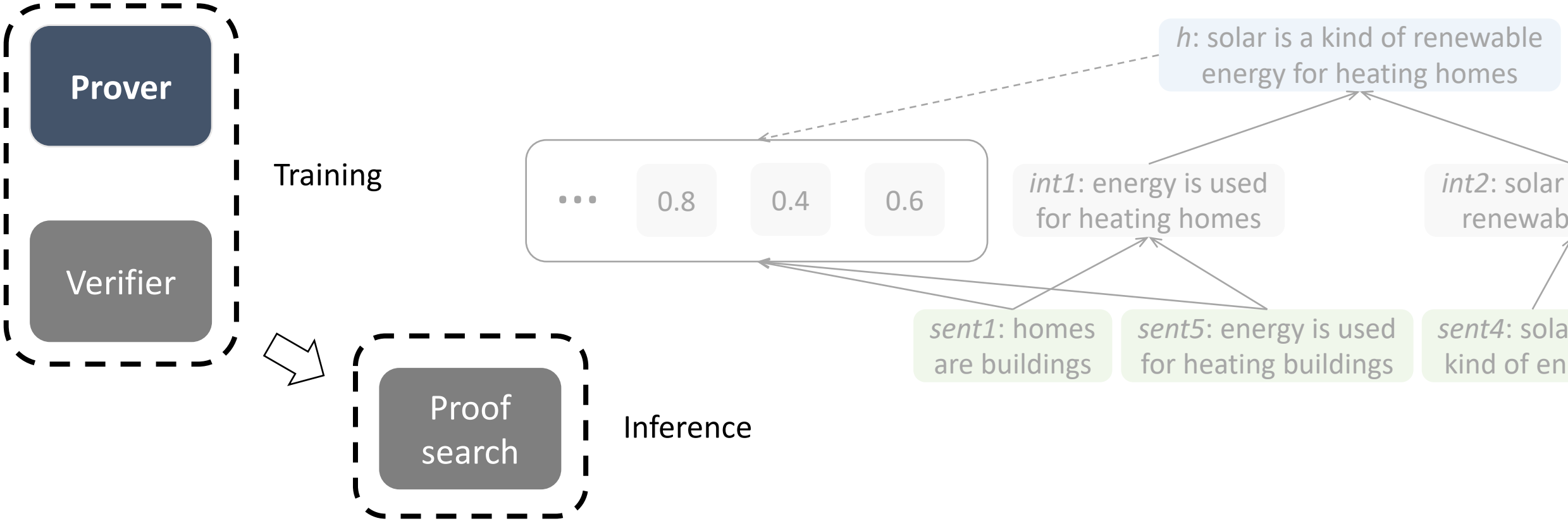

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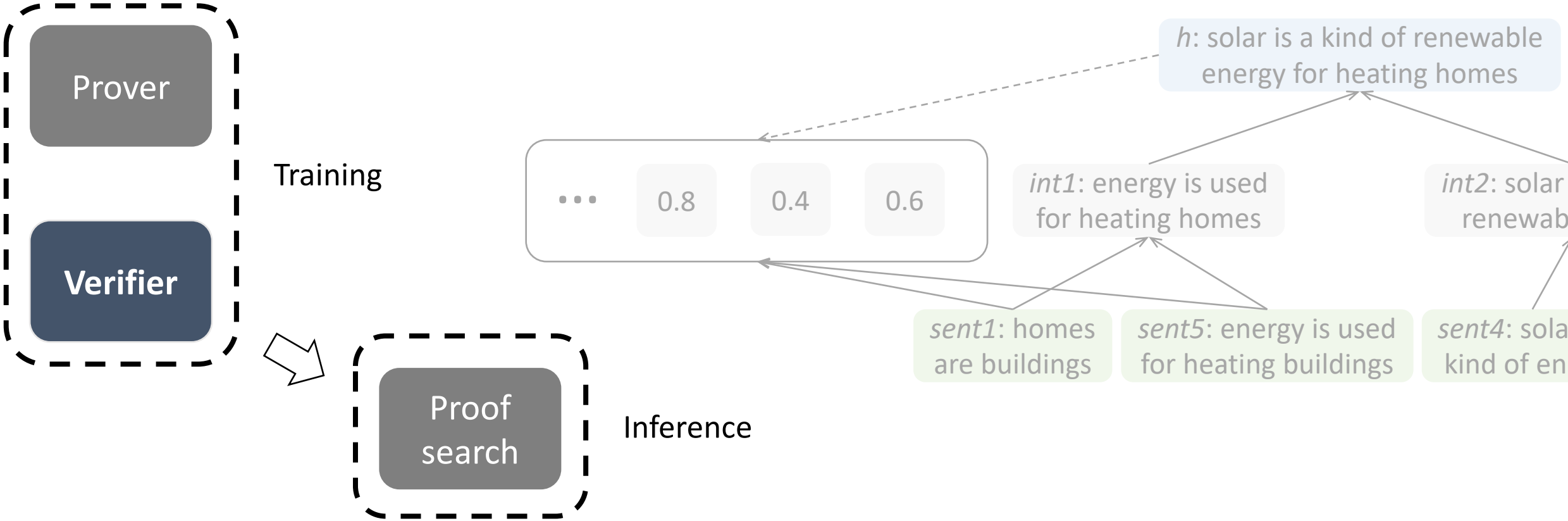

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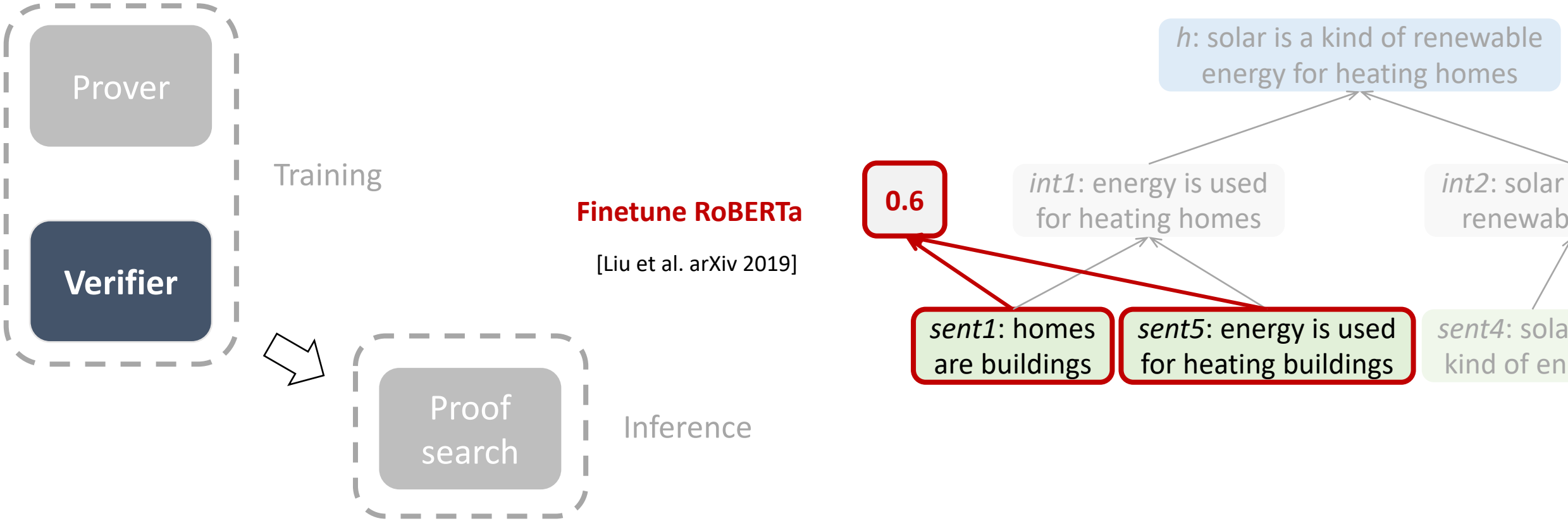
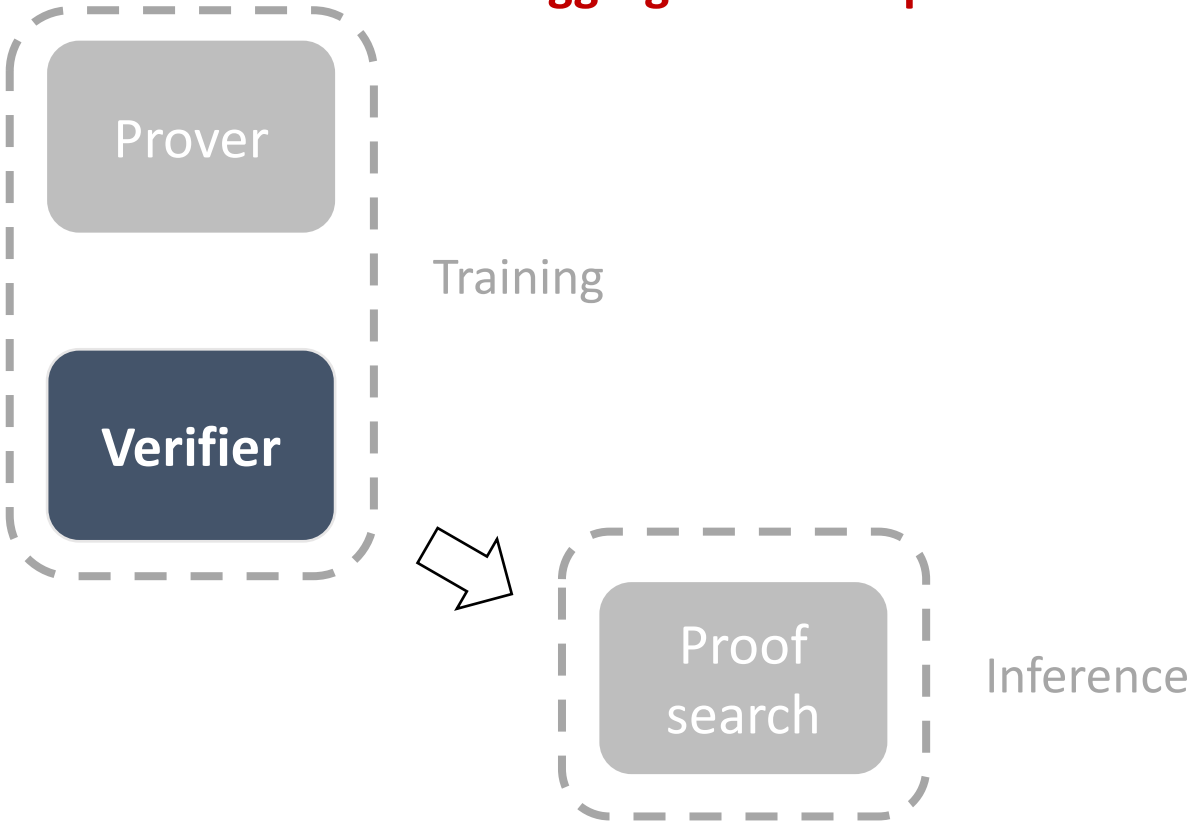
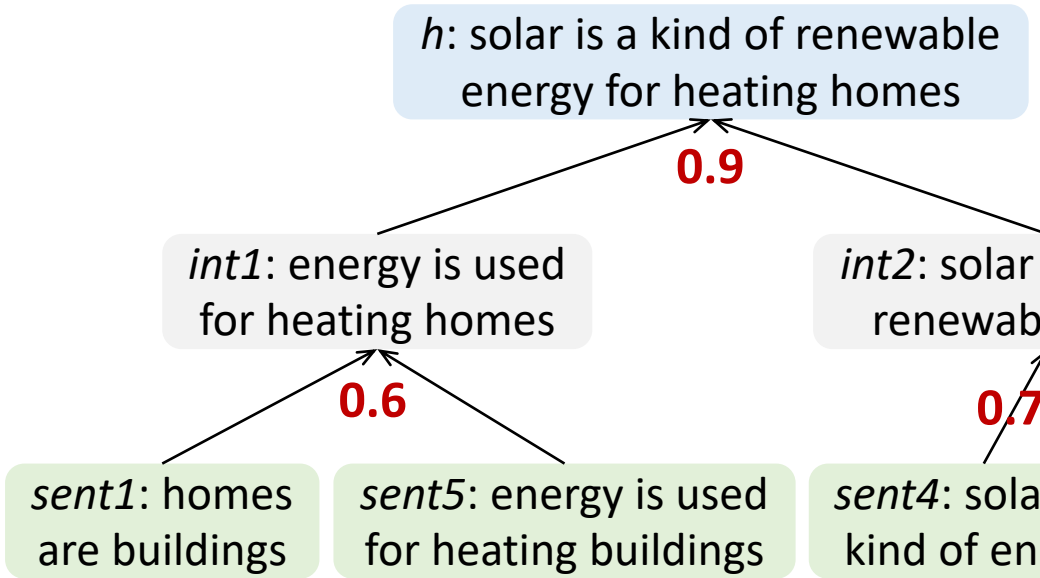


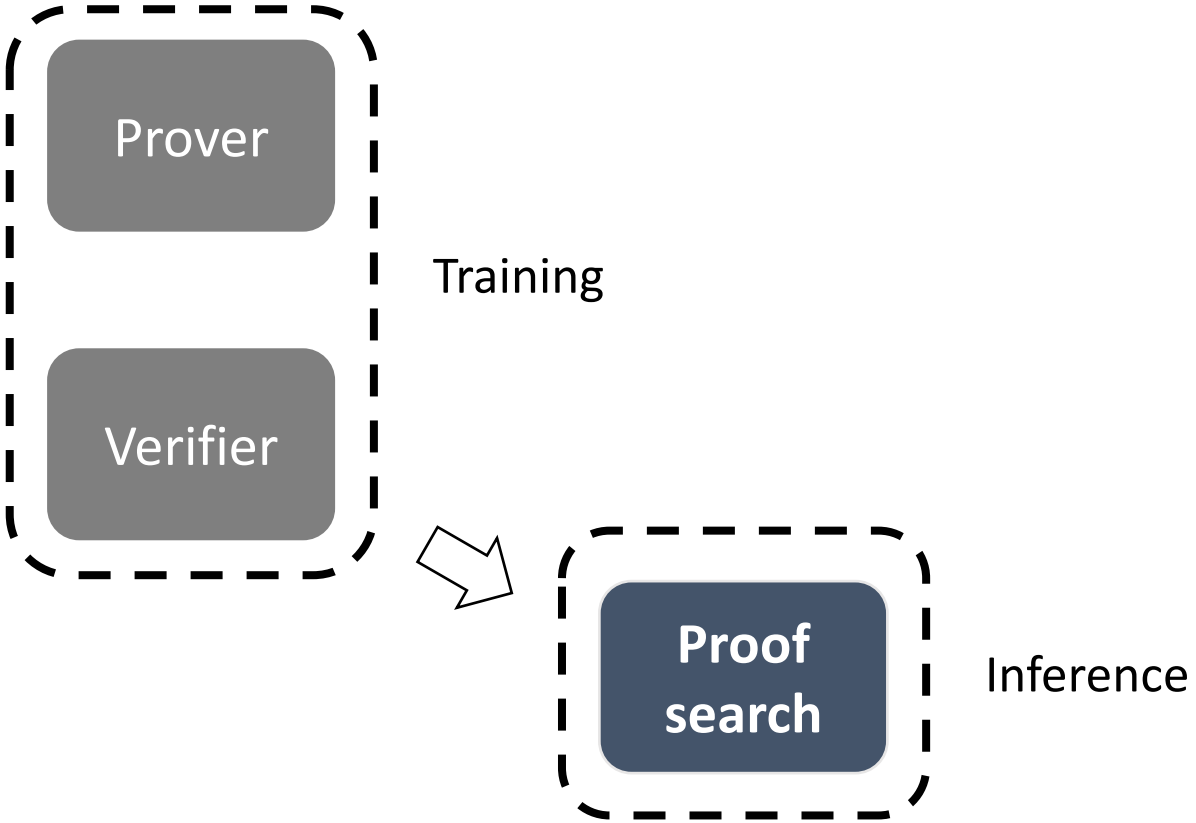
NLProofS: Natural Language Proof Search

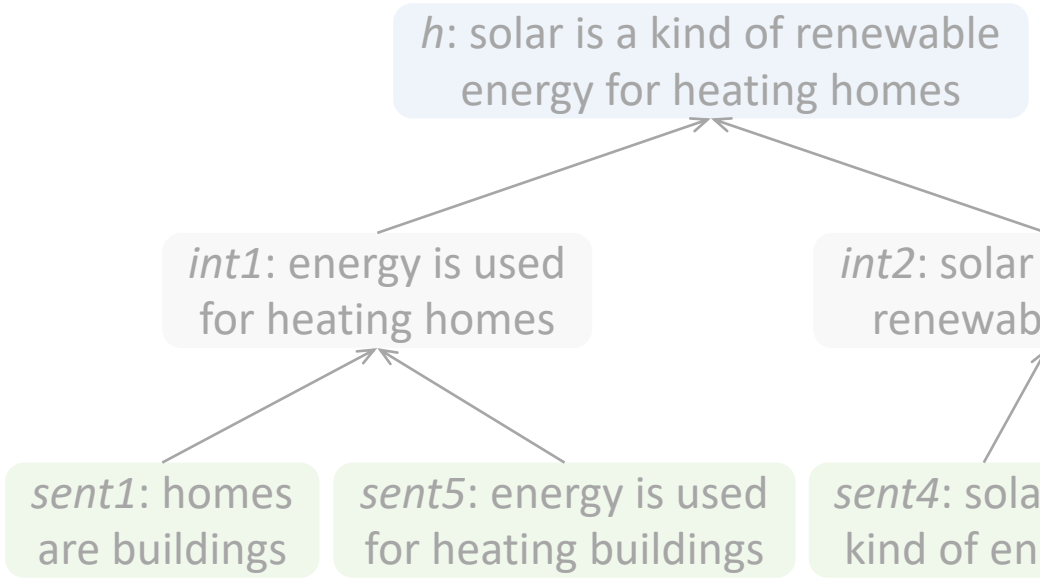
Aggregate the step scores to across the entire proof





NLProofS: Natural Language Proof Search





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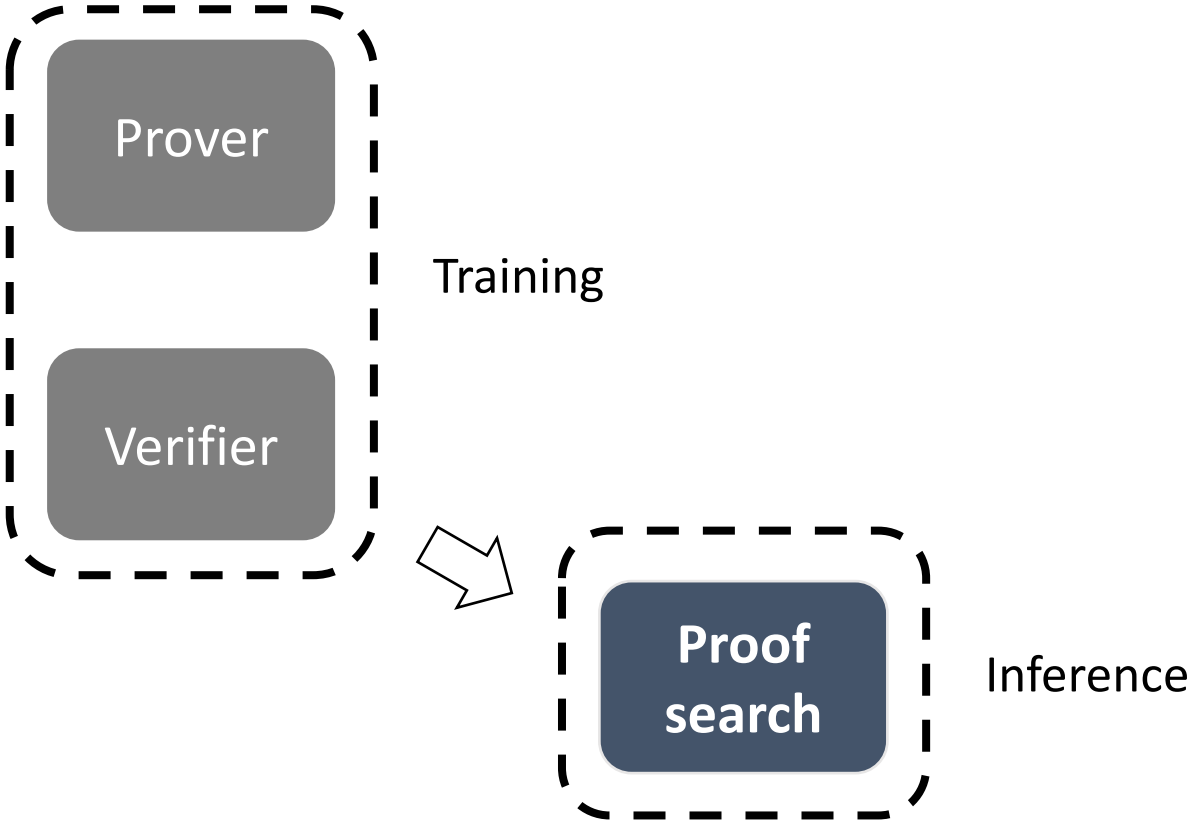

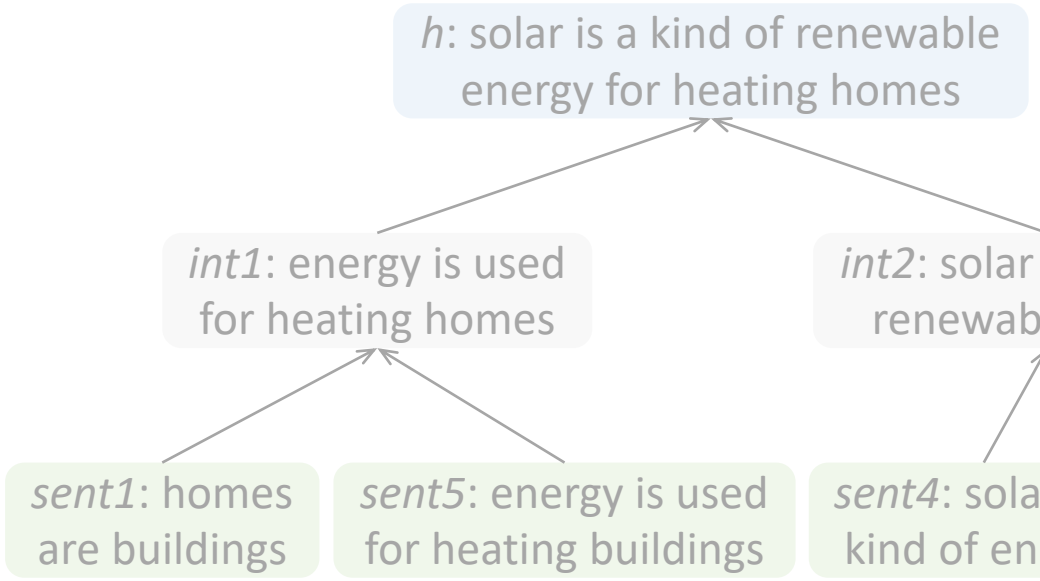

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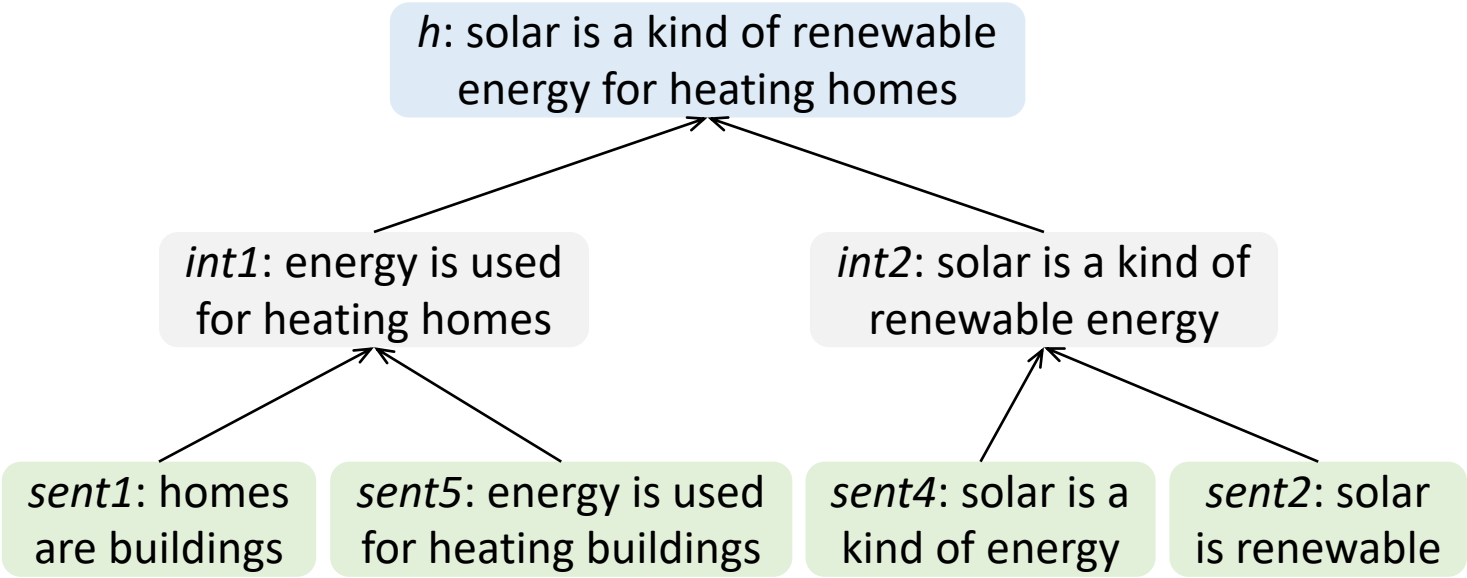
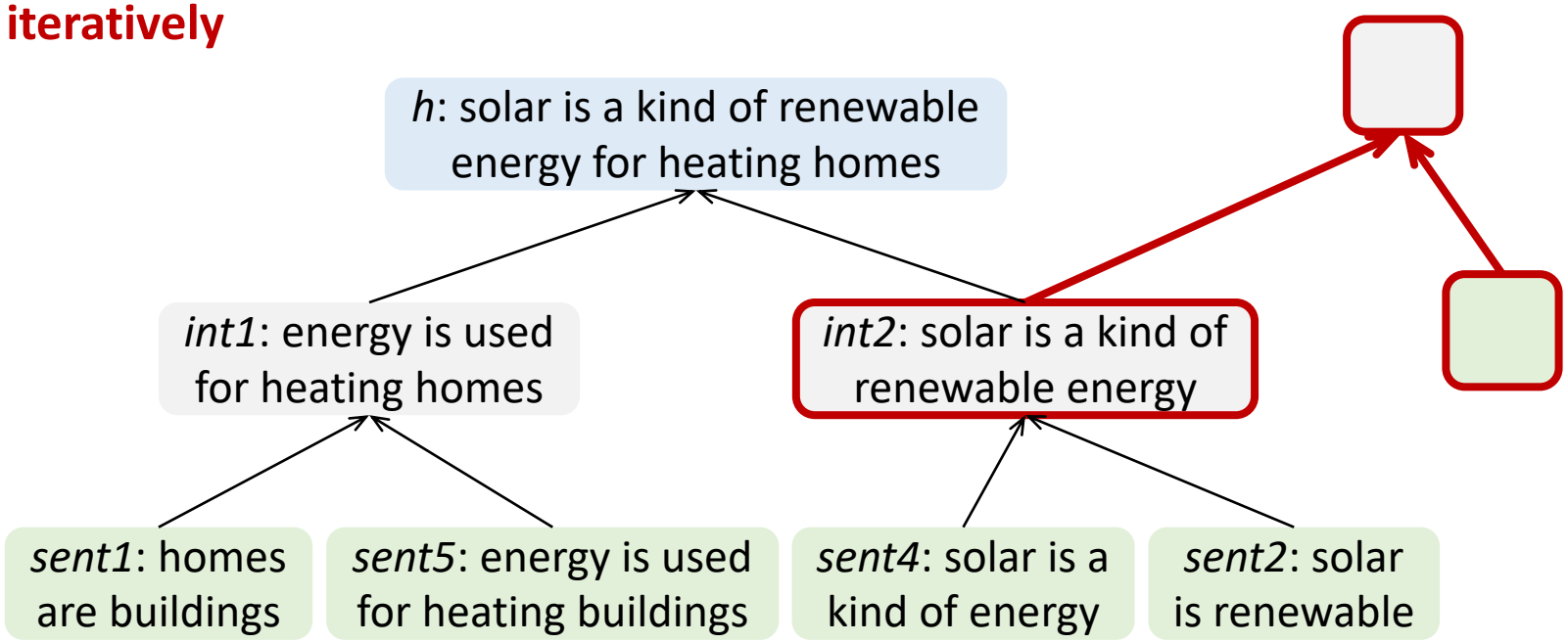


Proof search

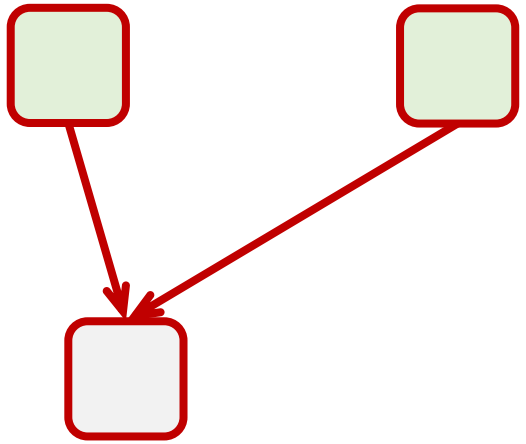
NLProofS: Natural Language Proof Search

1. Initialization: a proof generated by the prover alone
2. **Iteration: expand the graph iteratively**



Proof search

NLProofS: Natural Language Proof Search



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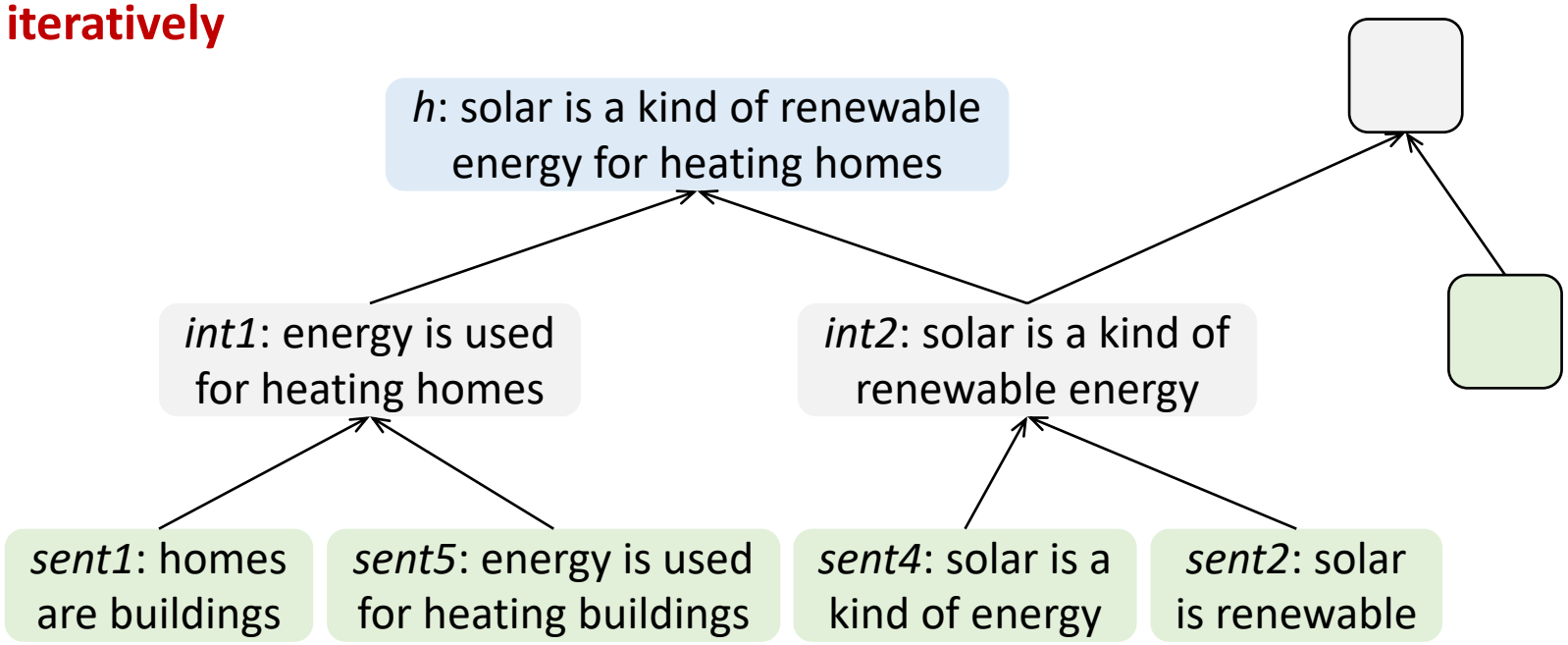

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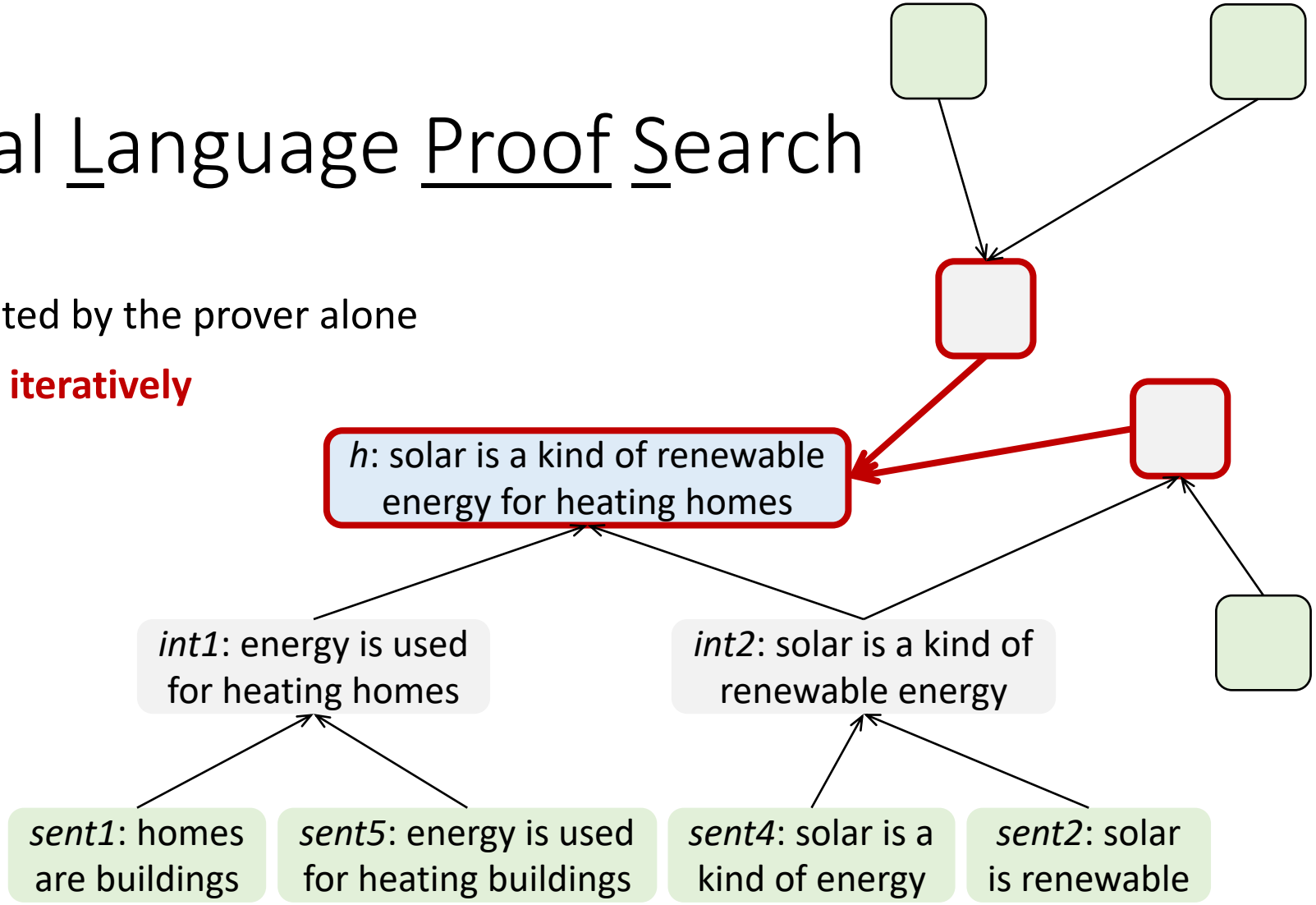
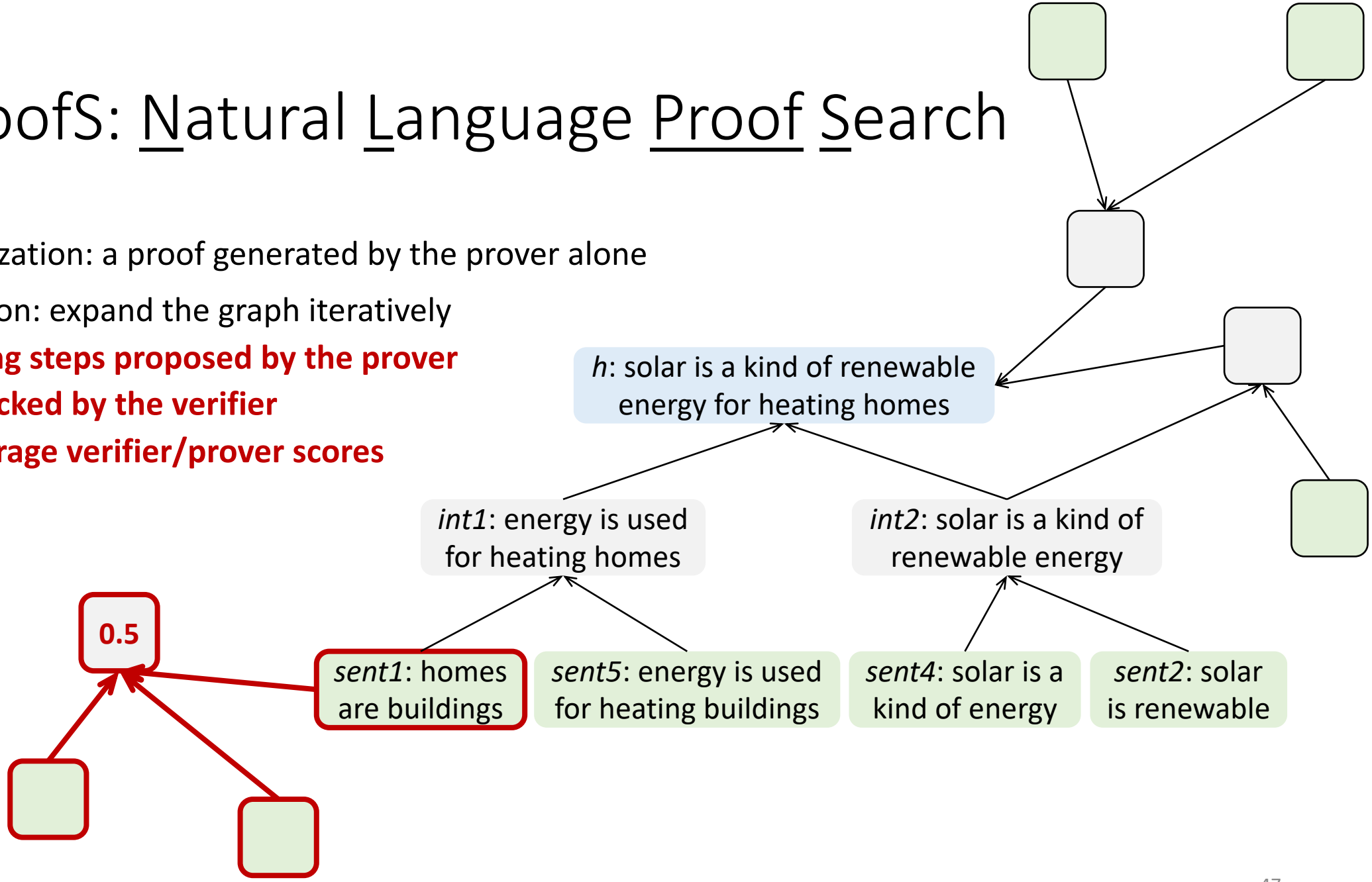


Proof search

NLProofS: Natural Language Proof Search

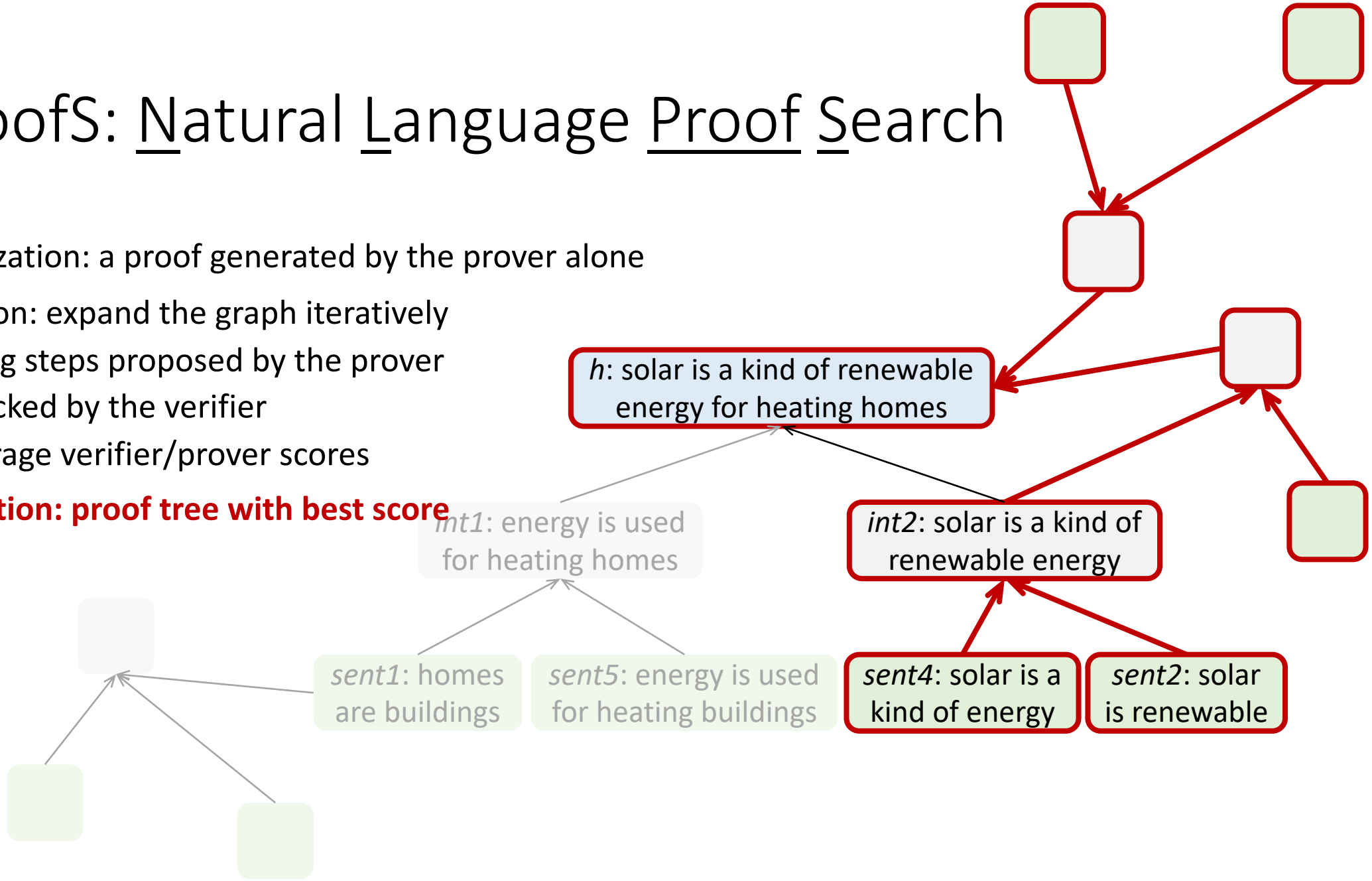
1. Initialization: a proof generated by the prover alone
2. Iteration: expand the graph iteratively
   - **Using steps proposed by the prover**
   - **Checked by the verifier**
   - **Average verifier/prover scores**



Proof search

NLProofS: Natural Language Proof Search

1. Initialization: a proof generated by the prover alone
2. Iteration: expand the graph iteratively
   - Using steps proposed by the prover
   - Checked by the verifier
   - Average verifier/prover scores
3. **Extraction: proof tree with best score**



Proof search

Experiments

- Evaluate on two benchmarks
  - **RuleTaker**: Simple, synthetic proofs [Tafjord et al. Findings of ACL 2021]
  - **EntailmentBank**: ~2K challenging, human-written proofs [Dalvi et al. EMNLP 2021]
- State-of-the-art results on both
- Ablations highlight the importance of the verifier

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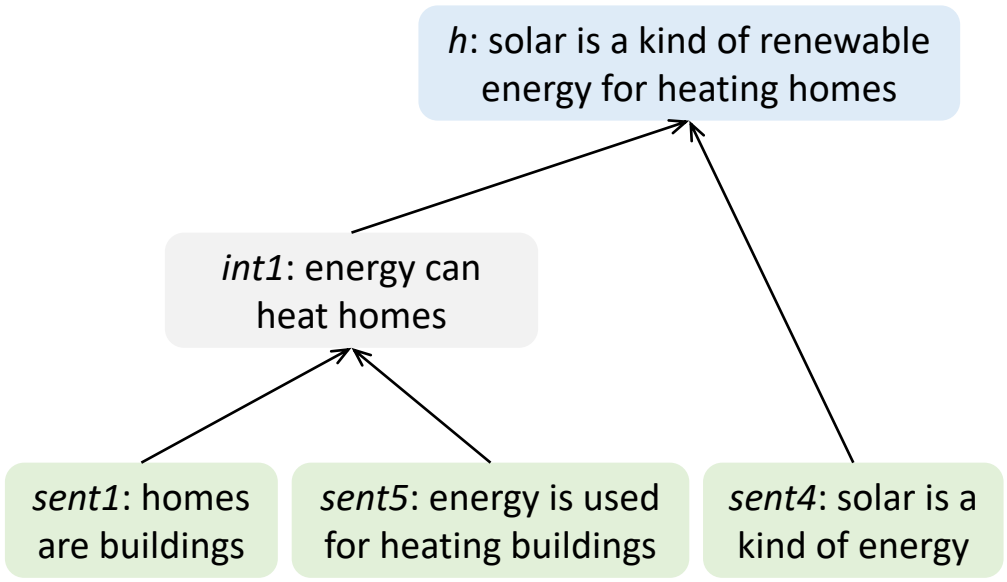
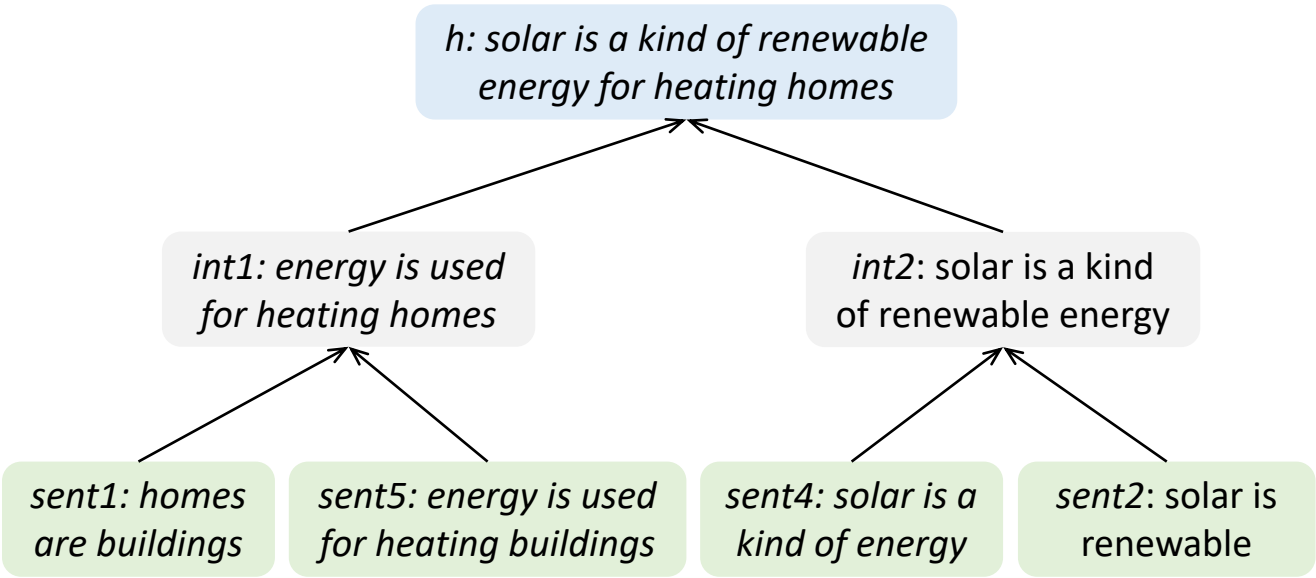
25 supporting facts, including distractors

[Dalvi et al. EMNLP 2021]

EntailmentBank: Evaluation Metrics



Predicted proof tree

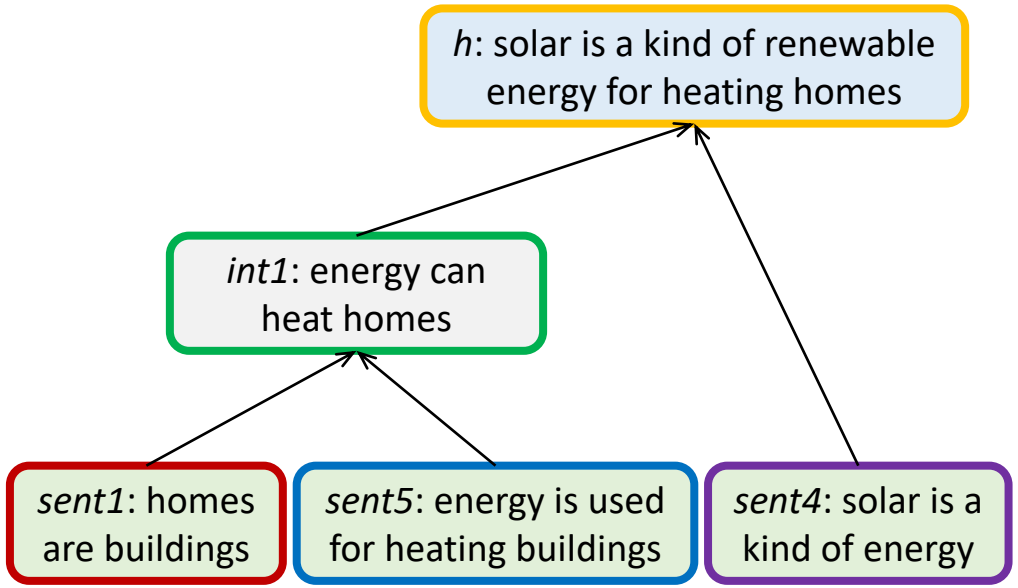


Ground truth proof tree

- EntailmentBank's four official metrics: Leaves, Steps, Intermediates, Overall

[Dalvi et al. EMNLP 2021]

EntailmentBank: Evaluation Metrics



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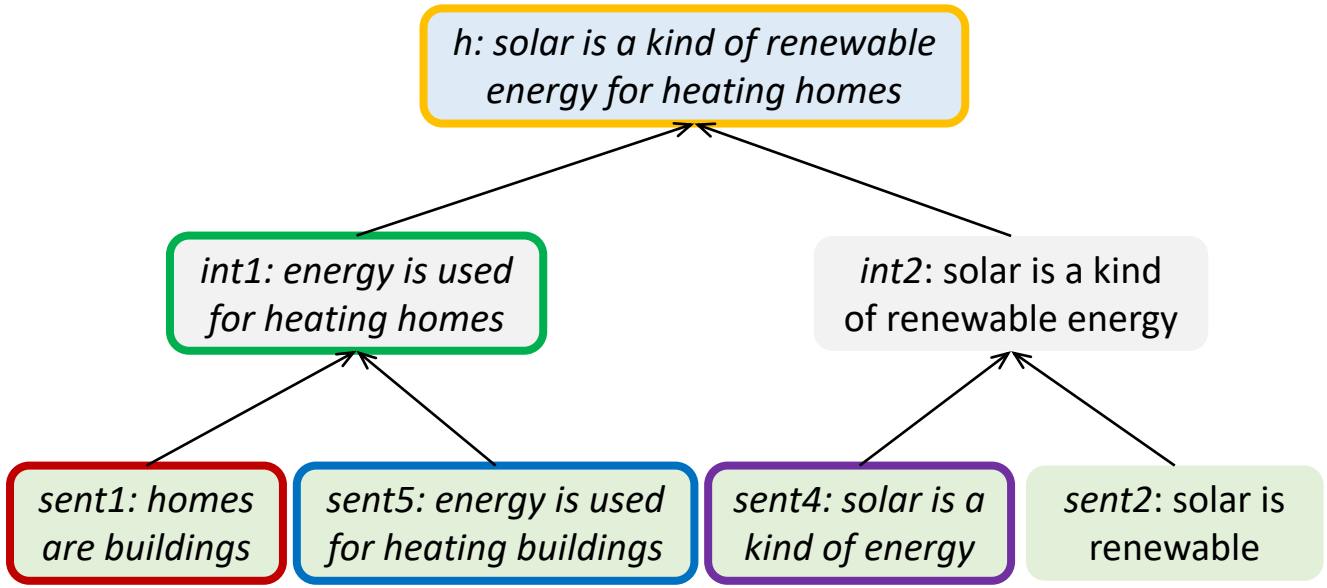

Ground truth proof tree

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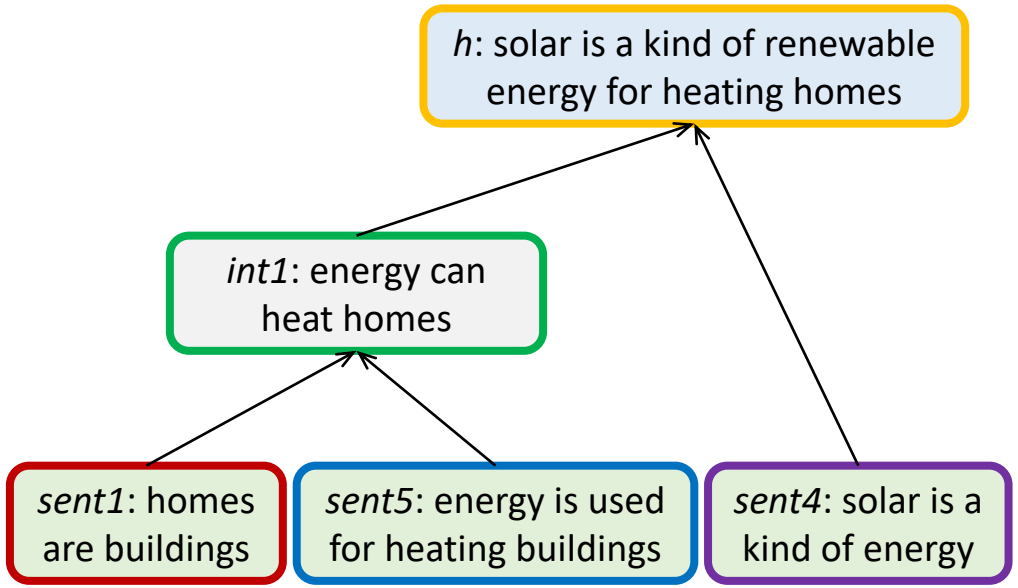
- **Based on heuristic matching between the nodes**

[Dalvi et al. EMNLP 2021]

EntailmentBank: Evaluation Metrics



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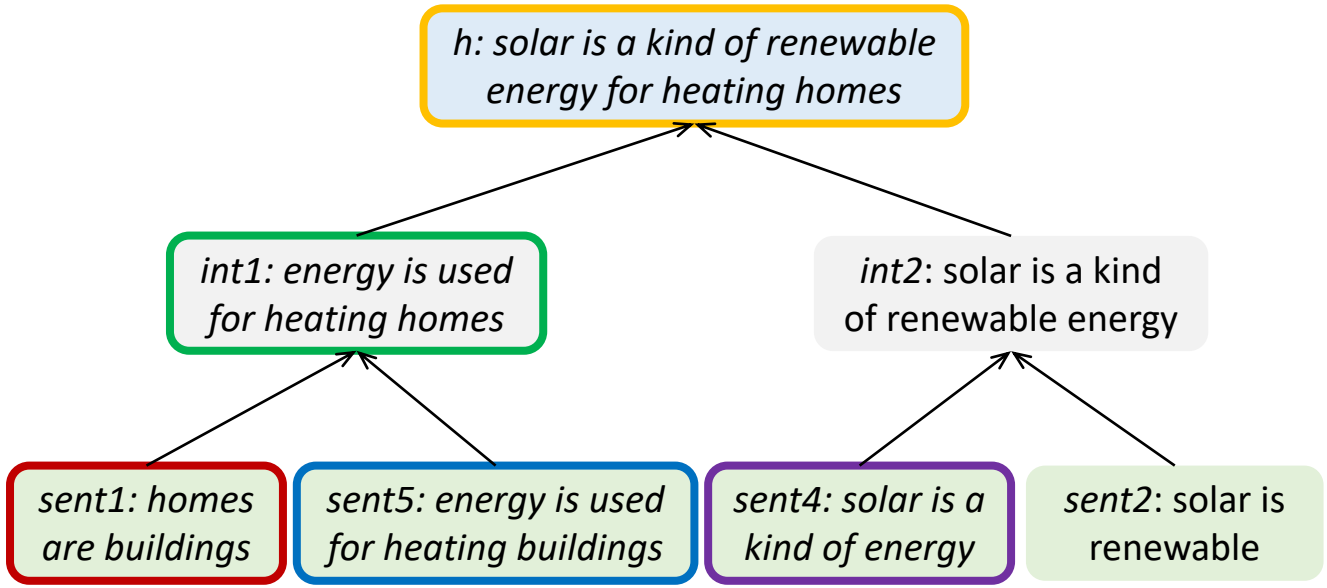
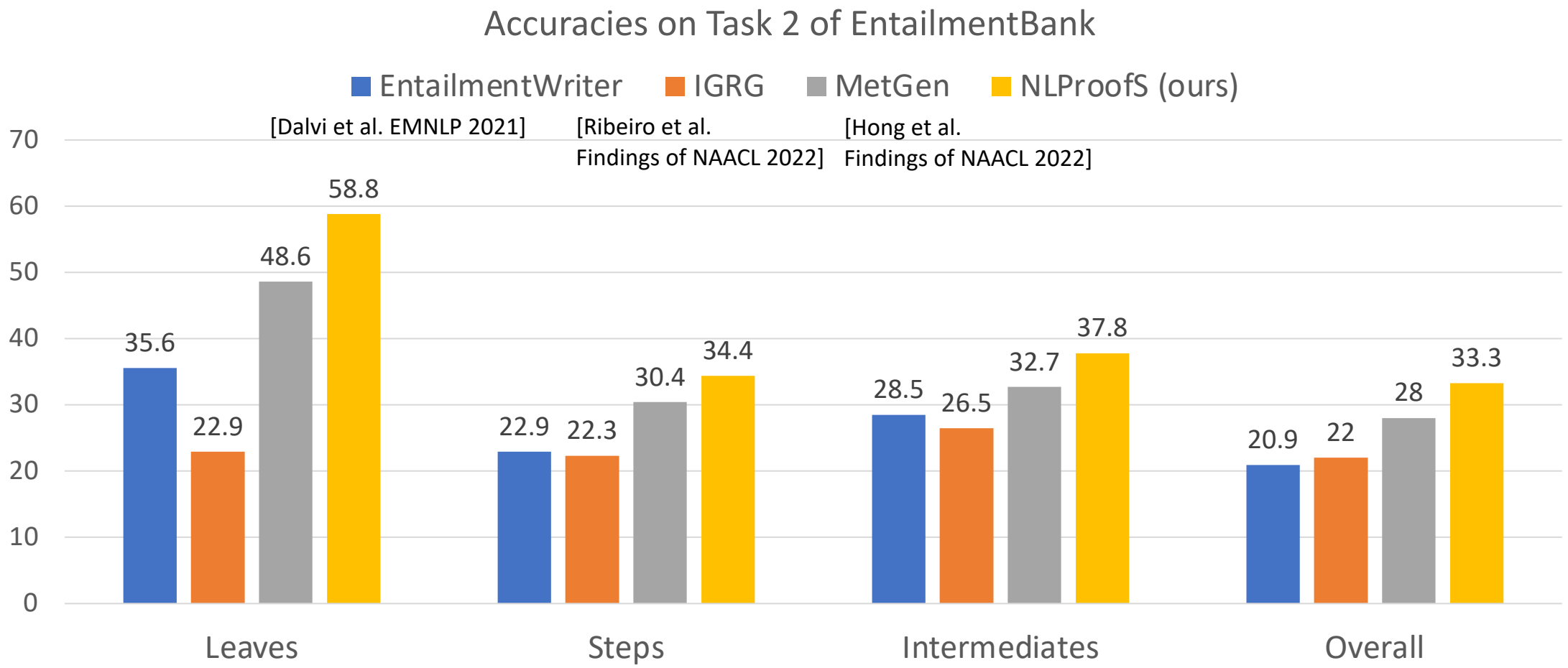


Ground truth proof tree

- EntailmentBank's four official metrics: Leaves, Steps, Intermediates, Overall
- Based on heuristic matching between the nodes
- **Limitations: Cannot handle correct predictions different from the ground truth**

State-of-the-art Performance on EntailmentBank



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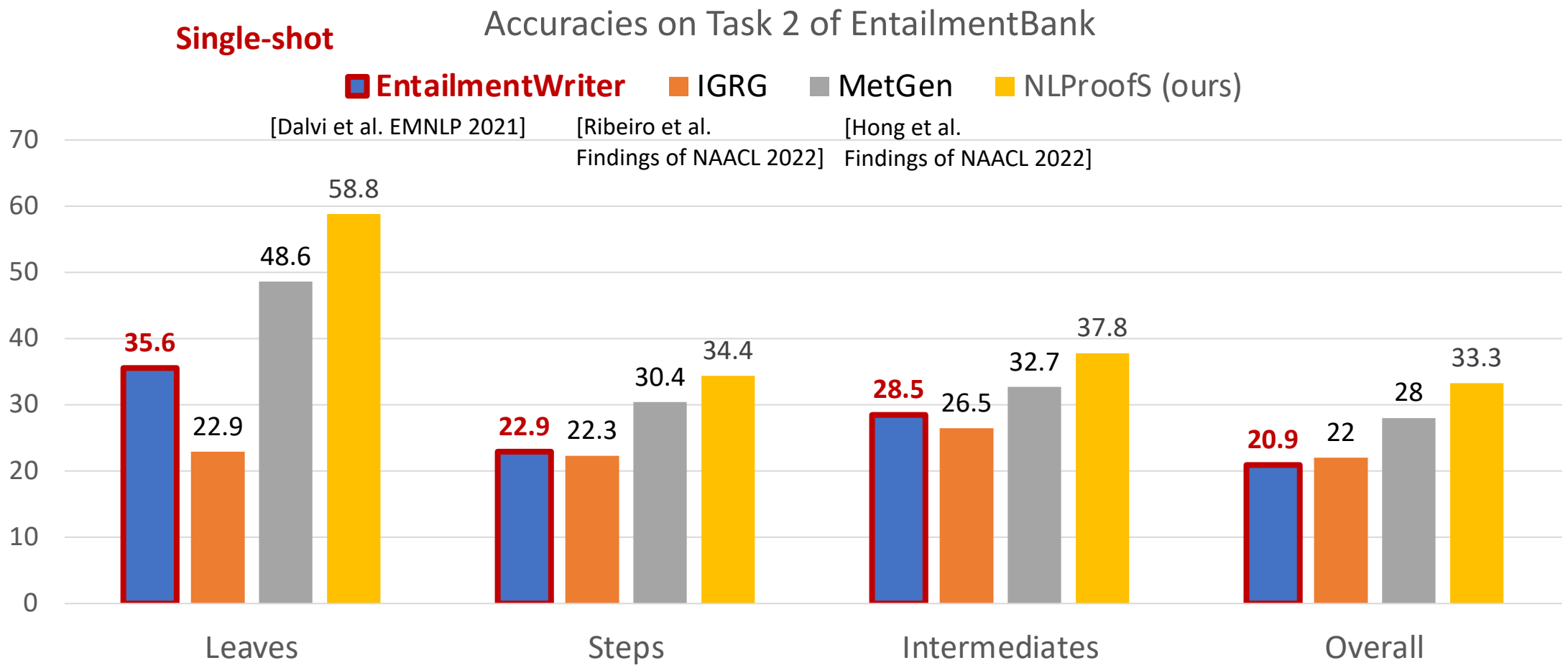

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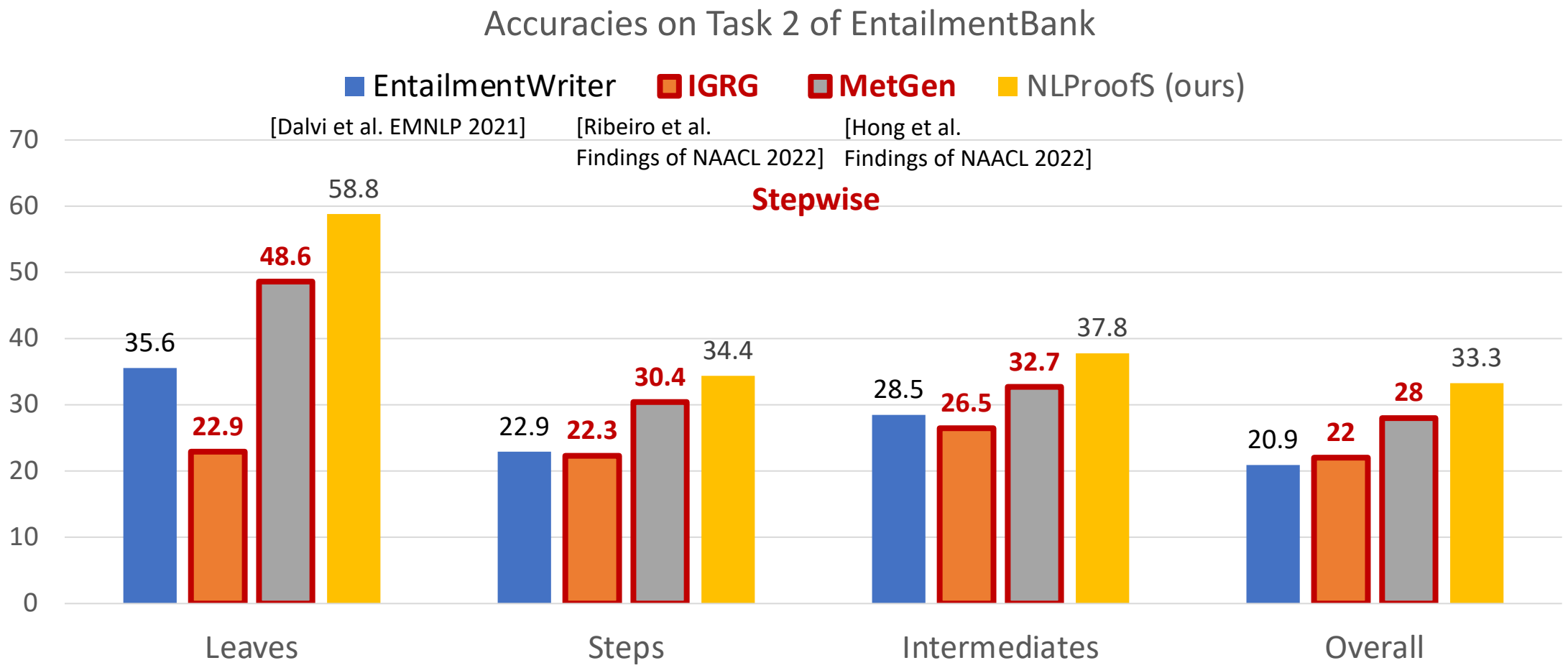

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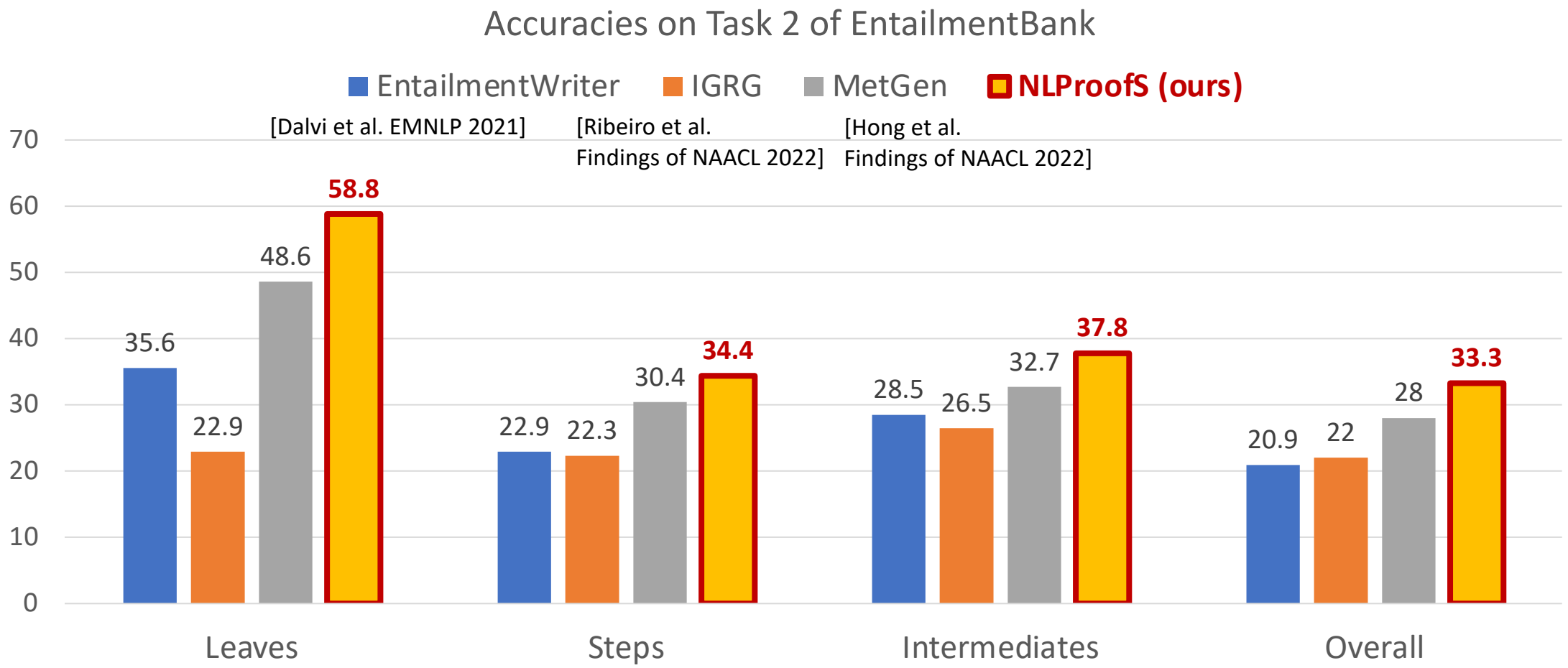

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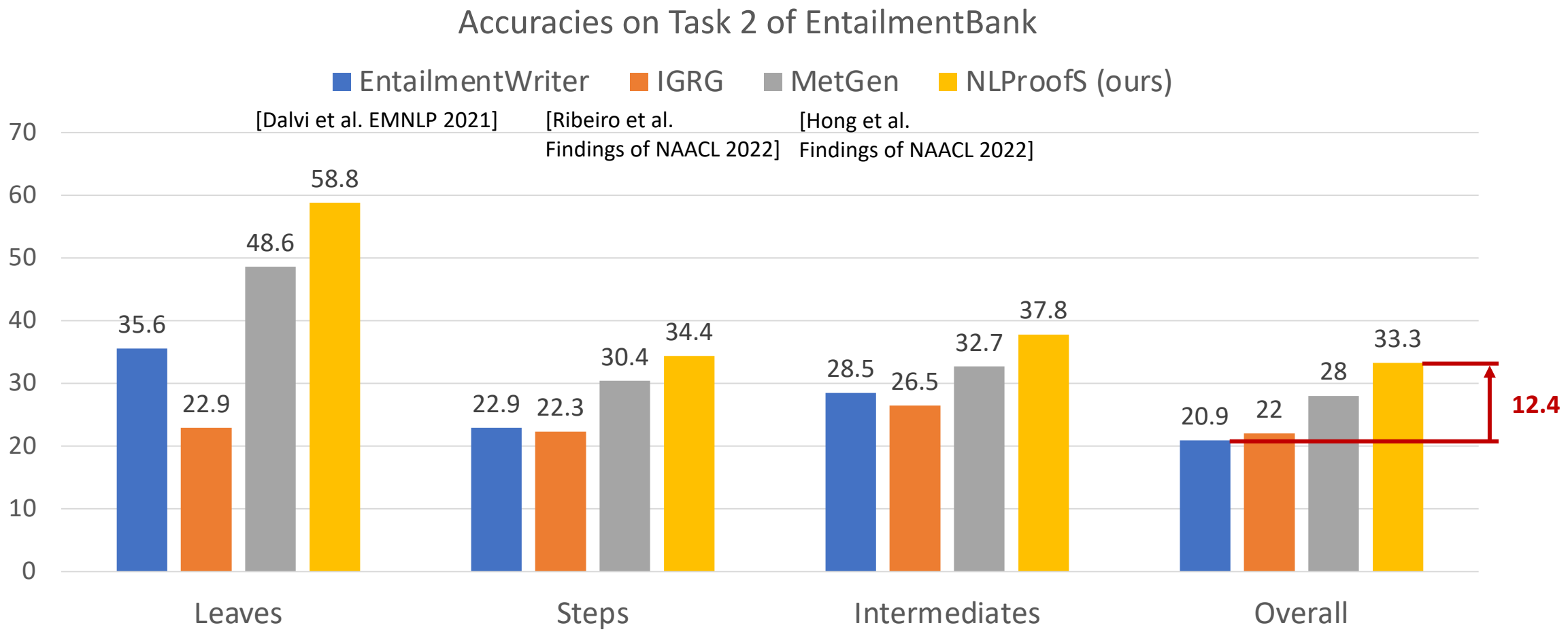

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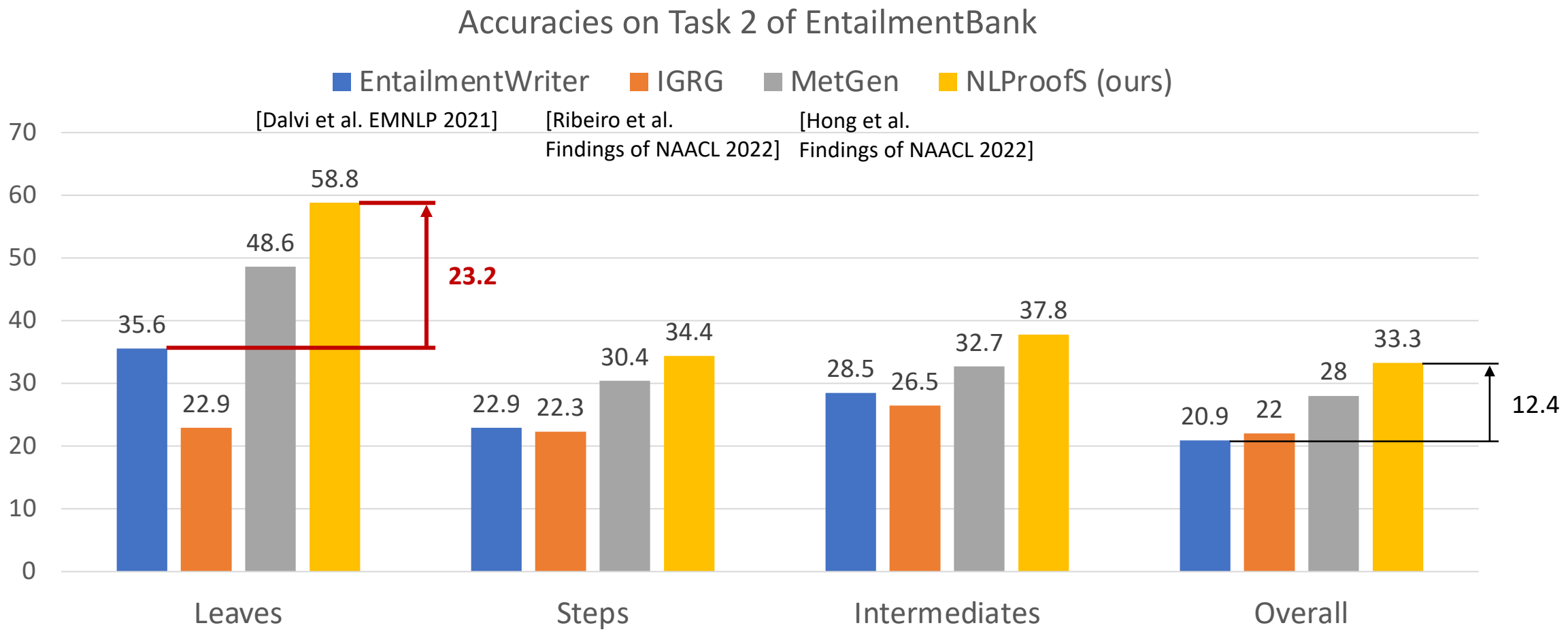

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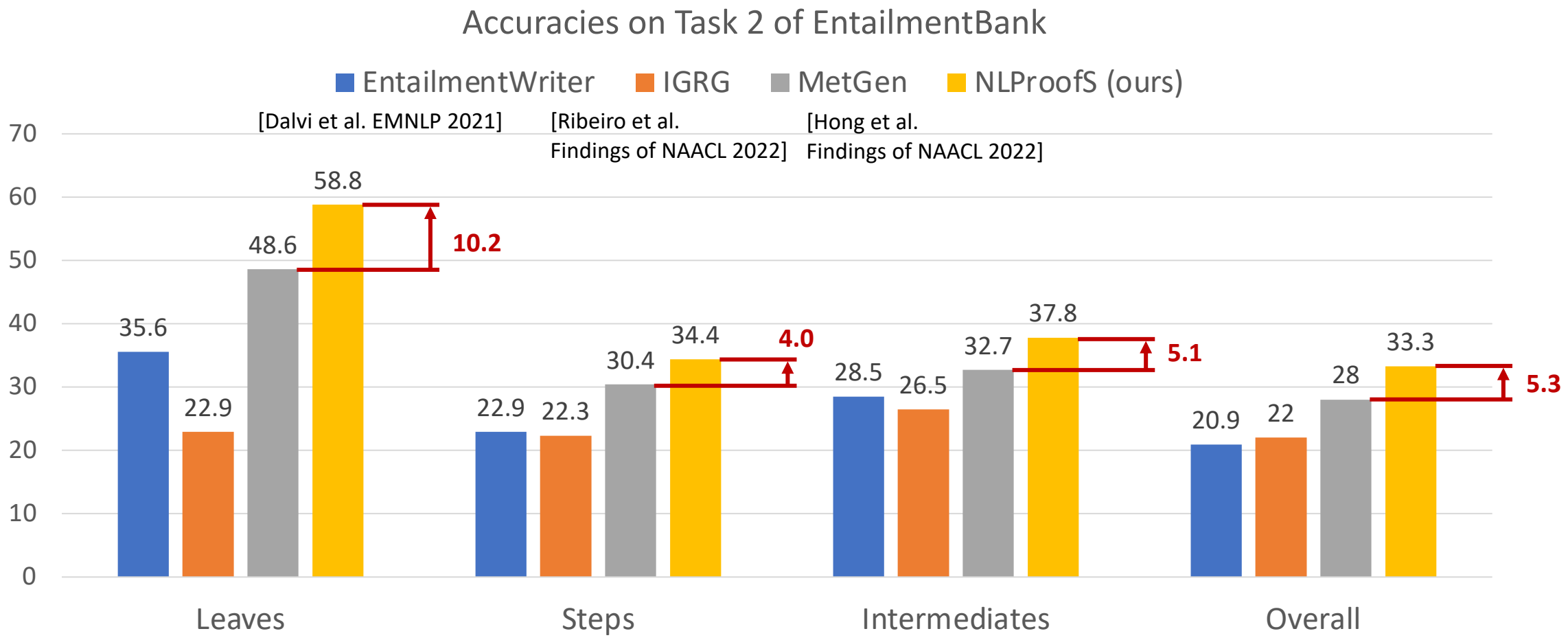
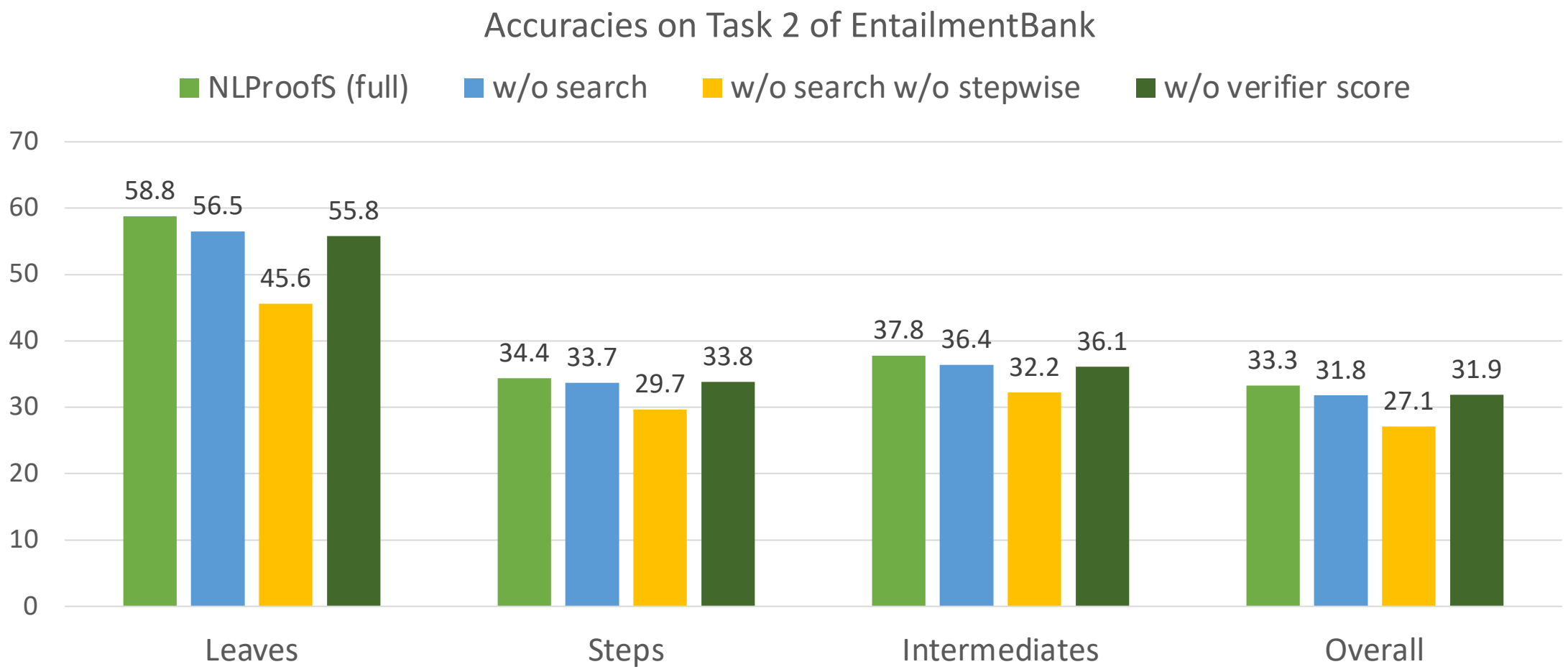
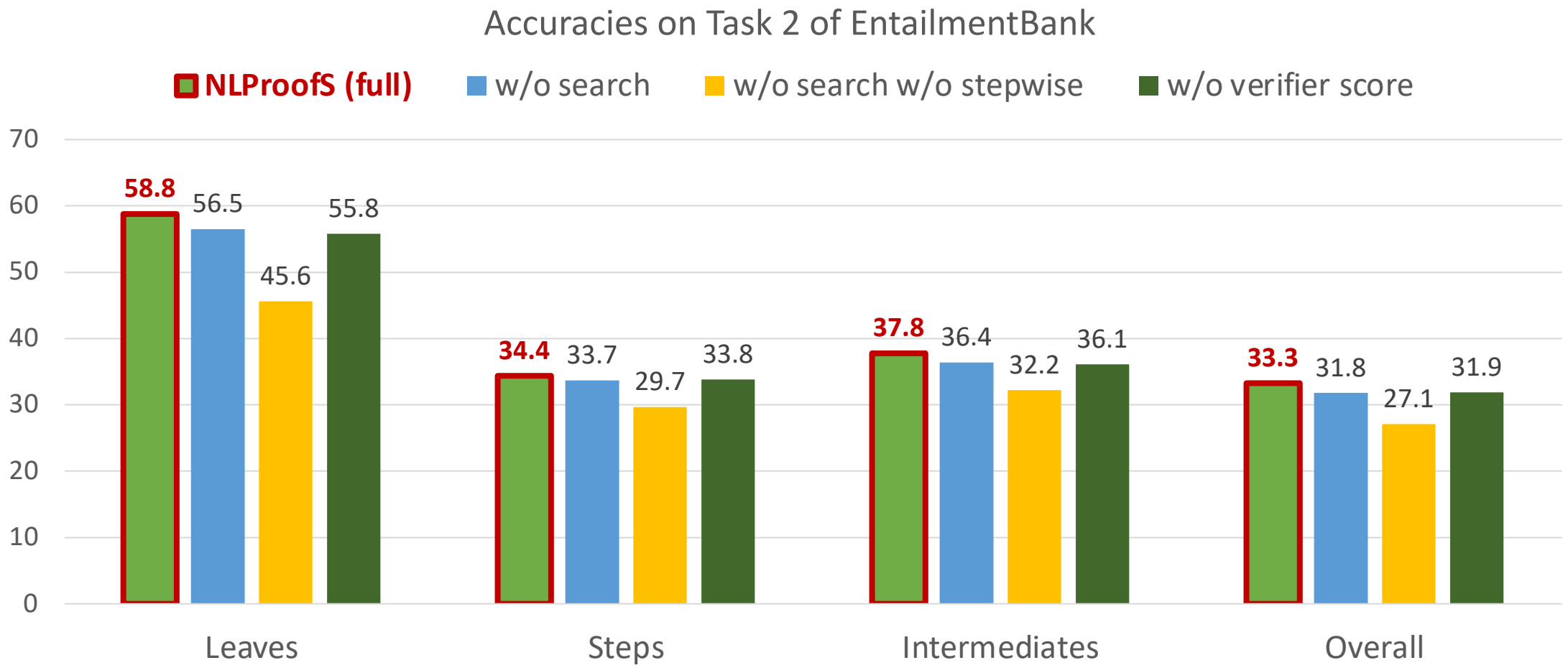


Ablations


Accuracies on Task 2 of EntailmentBank
NLProofS (full)
w/o search
w/o search w/o stepwise
w/o verifier score
70
60
50
40
30
20
10
0
58.8
56.5
45.6
55.8
34.4
33.7
29.7
33.8
37.8
36.4
32.2
36.1
33.3
31.8
27.1
31.9
Leaves
Steps
Intermediates
Overall

Ablations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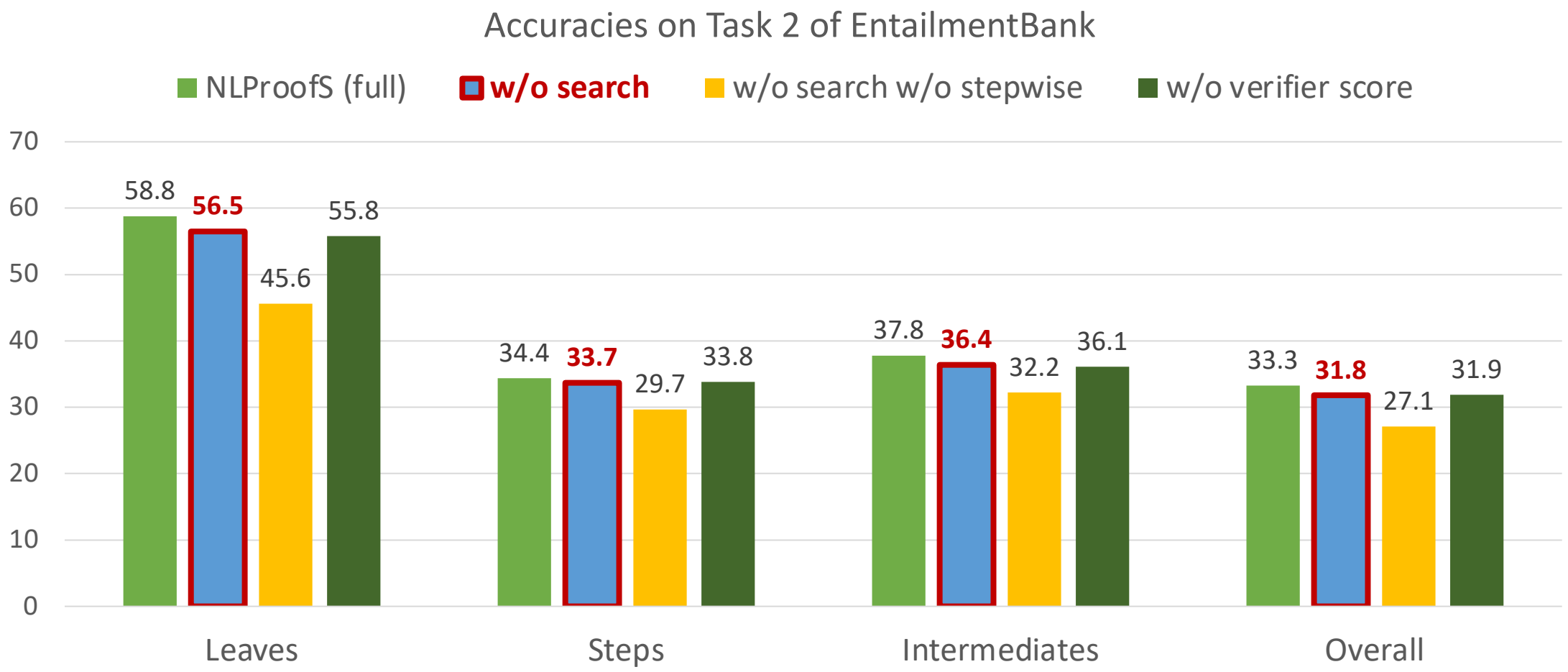
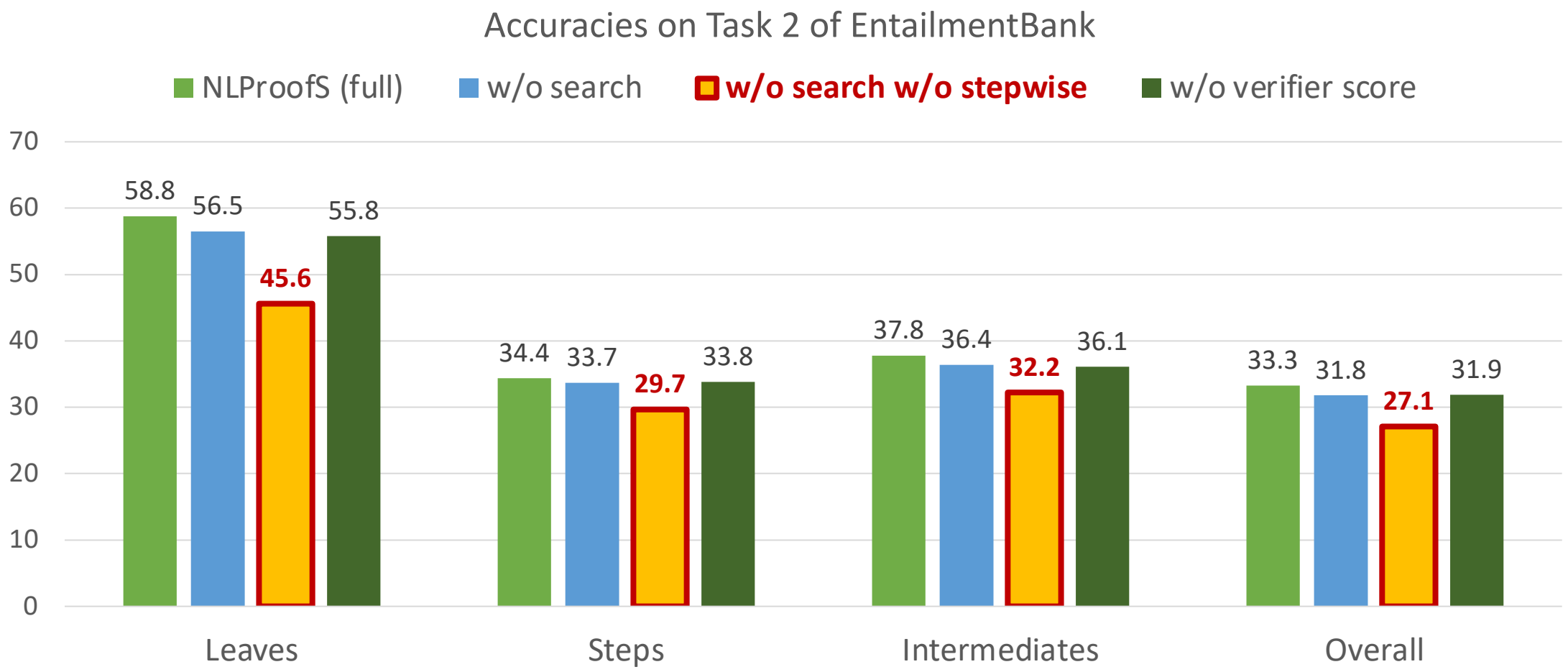
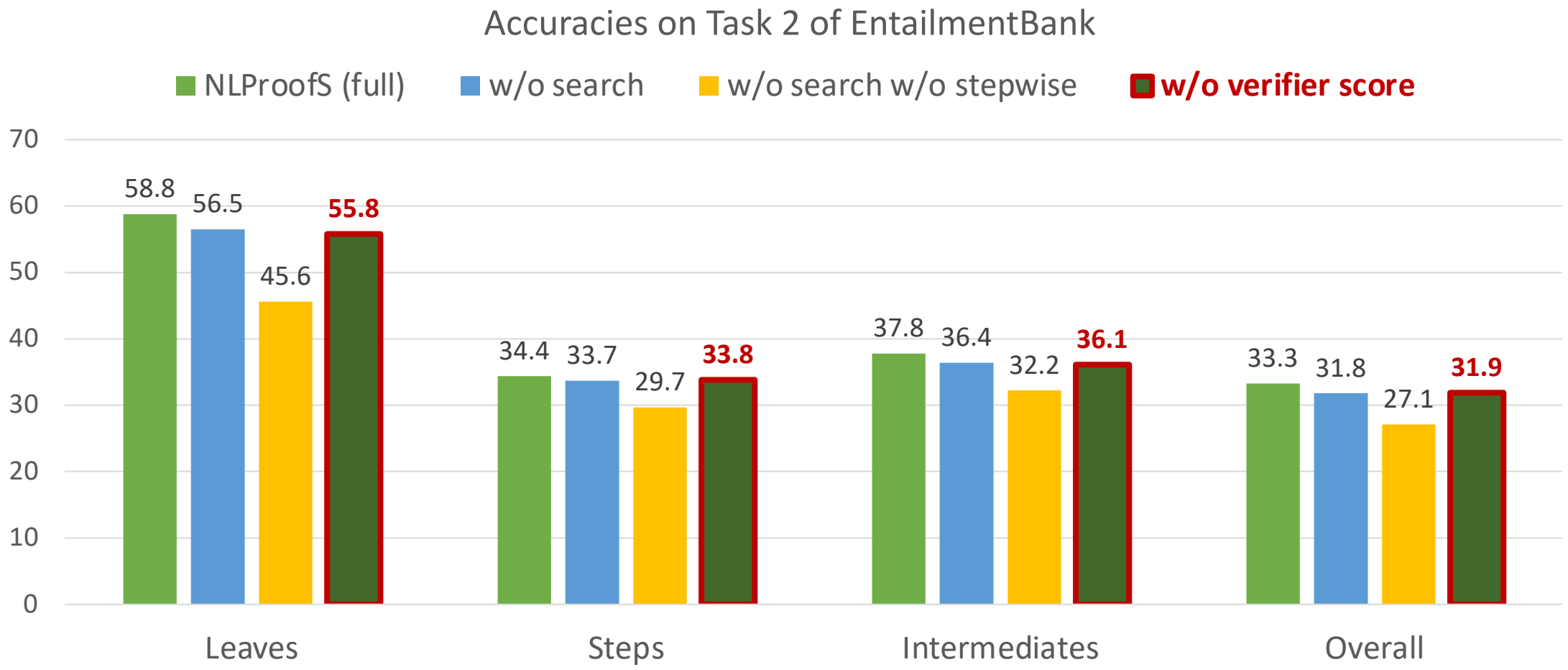
Verifier-Guided Proof Search Is Helpful



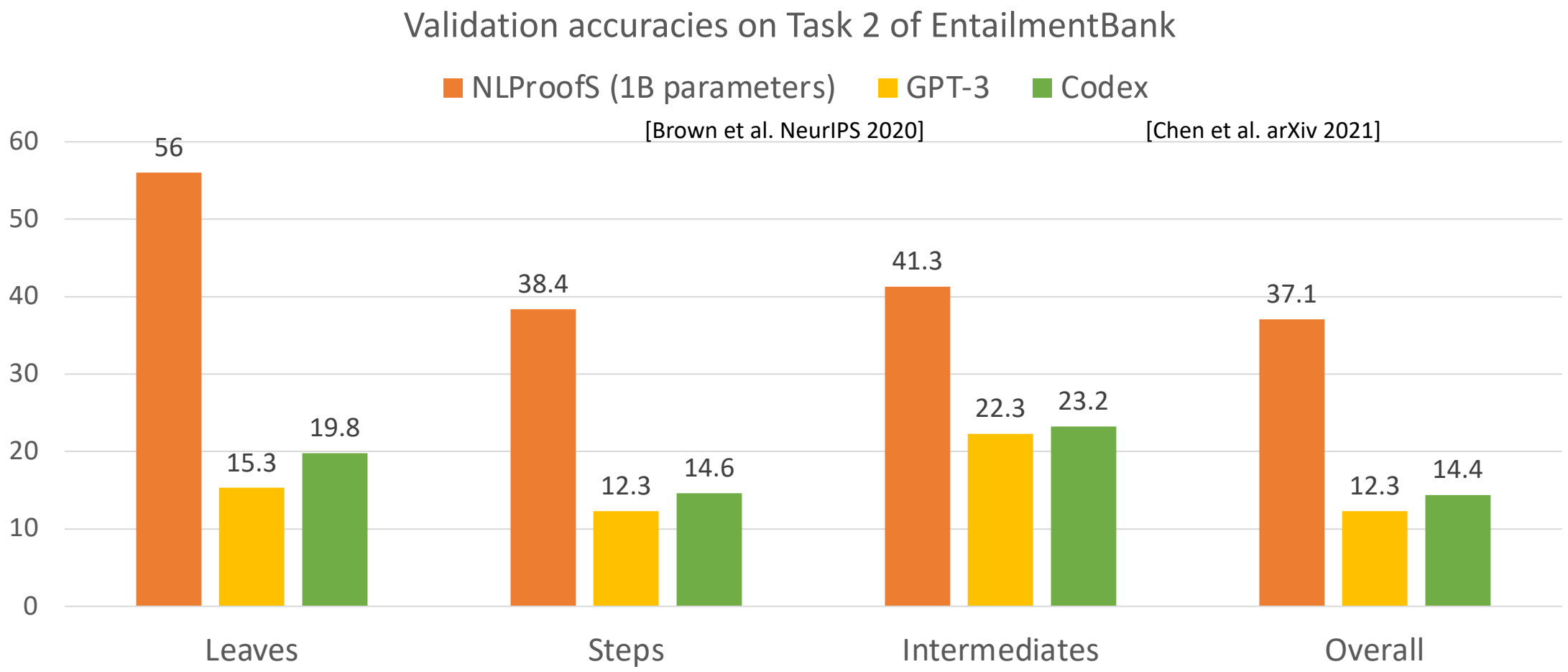
Stepwise Generation Is Helpful



The Verifier Is Necessary for Proof Search



Large Language Models w/ In-Context Learning



Key Takeaways

- The verifier is important
  - **Prevent hallucinated generations**

Key Takeaways

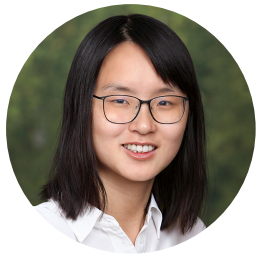
- The verifier is important
  - Prevent hallucinated generations
  - Also explored in other contexts, e.g., math word problems, code generation

[Cobbe et al. arXiv 2021] [Le and Wang et al. NeurIPS 2022]

Generating Natural Language Proofs with Verifier-Guided Search

Kaiyu Yang, Jia Deng, Danqi Chen

PRINCETON UNIVERSITY



https://github.com/princeton-nlp/NLProofS